



WSQ COURSE SLIDES

Agentic AI for Digital Marketing

n8n automation · SocialPost API · human oversight

Course Code: TGS-2025056988
WST-SNM-5042-1.1 · 16 hours

Version v2.1 · 29 August 2026



Digital Attendance (Mandatory)



AM attendance

Scan the SSG QR code before training.



PM attendance

Repeat after lunch and confirm submission.



Assessment attendance

Use the assessment QR before WA/PP.



Funding control

75% attendance and Competent result are required.

About the Trainer



Your Trainer

Template profile to be completed
by the trainer

Name

Industry role

Qualifications

**Relevant
experience**

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

Role

Principal trainer and courseware architect

Focus

AI, automation, cloud and applied digital transformation

Delivery

Mechanism-led training with auditable hands-on evidence

Practice

Human-supervised agent systems and responsible deployment

Let's Know Each Other

1

Role

What marketing decision do you own?

2

Current stack

Which channels and systems do you use?

3

Automation

What repetitive work should n8n coordinate?

4

Risk

What must never publish without human review?

Ground Rules

1

Participate

Ask, test and explain your evidence.

2

Respect

One conversation at a time.

3

Privacy

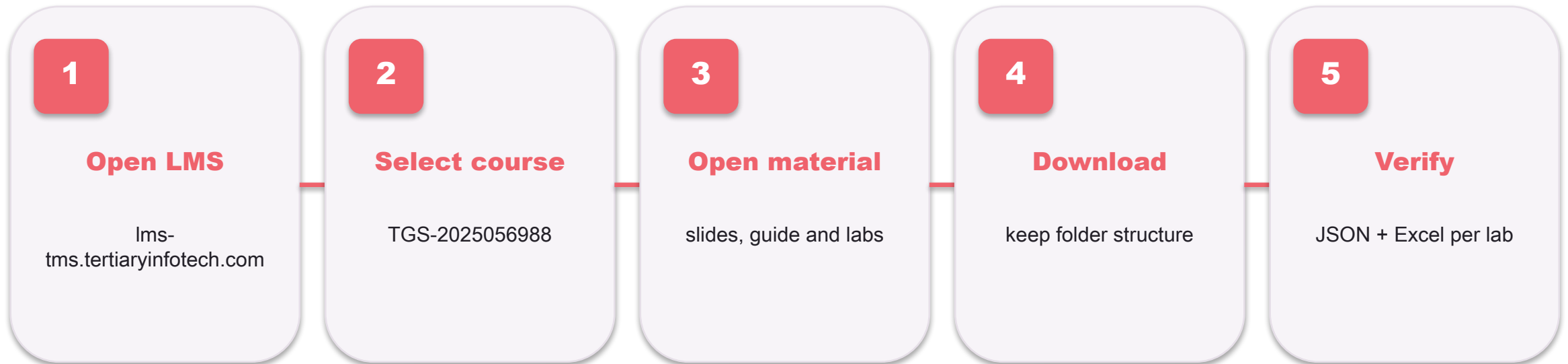
Use only synthetic data and sandbox accounts.

4

Safety

Keep live publisher nodes disabled until authorised.

Download Course Material



Use the Learner Guide for detailed procedures

16 HOURS

Course Schedule — 2 Days

AM

Day 1 AM

Research intelligence and campaign inputs.

PM

Day 1 PM

Strategy, KPI/ROI and content contracts.

AM

Day 2 AM

Human oversight and publishing.

PM

Day 2 PM

Analytics, optimisation and assessment.

Learning Outcomes

1

LO1

Research, strategy, objectives, KPIs and ROI.

2

LO2

Evaluate ROI and synthesise strategy models.

3

LO3

Lead an integrated presence with human oversight.

✓

Evidence

15 connected n8n labs and a run ledger.

Skills Framework



TSC

Digital Marketing · Level 5.



Knowledge

K1–K5: frameworks, analytics, models, ROI, channels.



Ability

A1–A5: KPIs, goals, ROI, integration, presence.



Assessment

Open-ended WA + hands-on PP.

Knowledge Criteria — K1 to K5

K1

Frameworks for designing an integrated digital marketing strategy

K2

Techniques to draw insights from marketing analytics

K3

Integrated Digital Marketing strategy and models

K4

Methods for analysis of digital marketing return on investment

K5

Developments in emerging marketing channels

Ability Criteria — A1 to A5

A1

Formulate KPIs for digital marketing channels

A2

Define goals and objectives of a digital marketing strategy

A3

Evaluate ROI for online customer-acquisition tools and channels

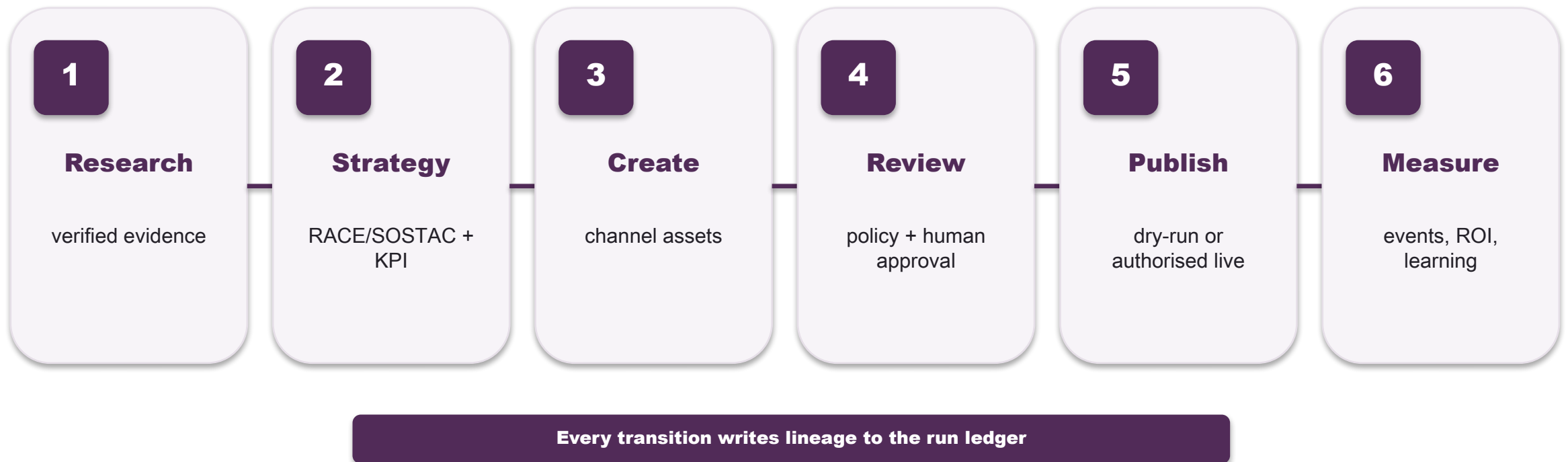
A4

Integrate digital marketing with the overall marketing strategy

A5

Lead development of a seamless online presence over web, social and mobile

End-to-End Marketing Operating System



Human Oversight Boundary



Failure mode

An agent can generate, score and recommend—but must not approve its own output, expand scope or mutate a live budget.



Control and recovery

n8n pauses consequential branches, binds approval to a payload hash, records the reviewer and routes changed payloads back to review.

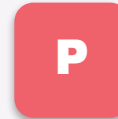
Automation stops safely; a named owner decides the next action.

Briefing for Assessment



WA/SAQ

60 minutes; open-ended knowledge questions.



PP

60 minutes; practical workflow evidence.



Submission

Upload completed answers to the LMS.



Conduct

No discussion, recording or script sharing.

Assessment



Knowledge

Explain mechanisms, formulas, controls and evidence.



Ability

Build and verify a supervised n8n stage.



Judgement

Identify risk and justify human oversight.



Result

Competent / Not Yet Competent.

Assessment Flow



Question papers only are learner-facing

TOPIC 01

Research Intelligence & Campaign Inputs

Evidence collection · source quality · audience signals · opportunity scoring

LO1 · K1-K2 · A1-A2

Architecture · runtime · contract · controls · evidence



Research contract

M

Mechanism

A structured brief defines market, audience, decision, constraints and evidence threshold.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Evidence lineage

M

Mechanism

Every insight retains source URL, retrieval time, publisher, claim and confidence.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Signal scoring

M

Mechanism

Relevance × reliability × recency produces an auditable priority score.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Audience synthesis

M

Mechanism

Jobs, pains, triggers and objections are derived from evidence—not invented personas.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Agent boundary

M

Mechanism

n8n orchestrates tools and state; a human owns research acceptance and risk.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Data minimisation

M

Mechanism

Collect only fields required for a marketing decision; exclude sensitive personal data.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

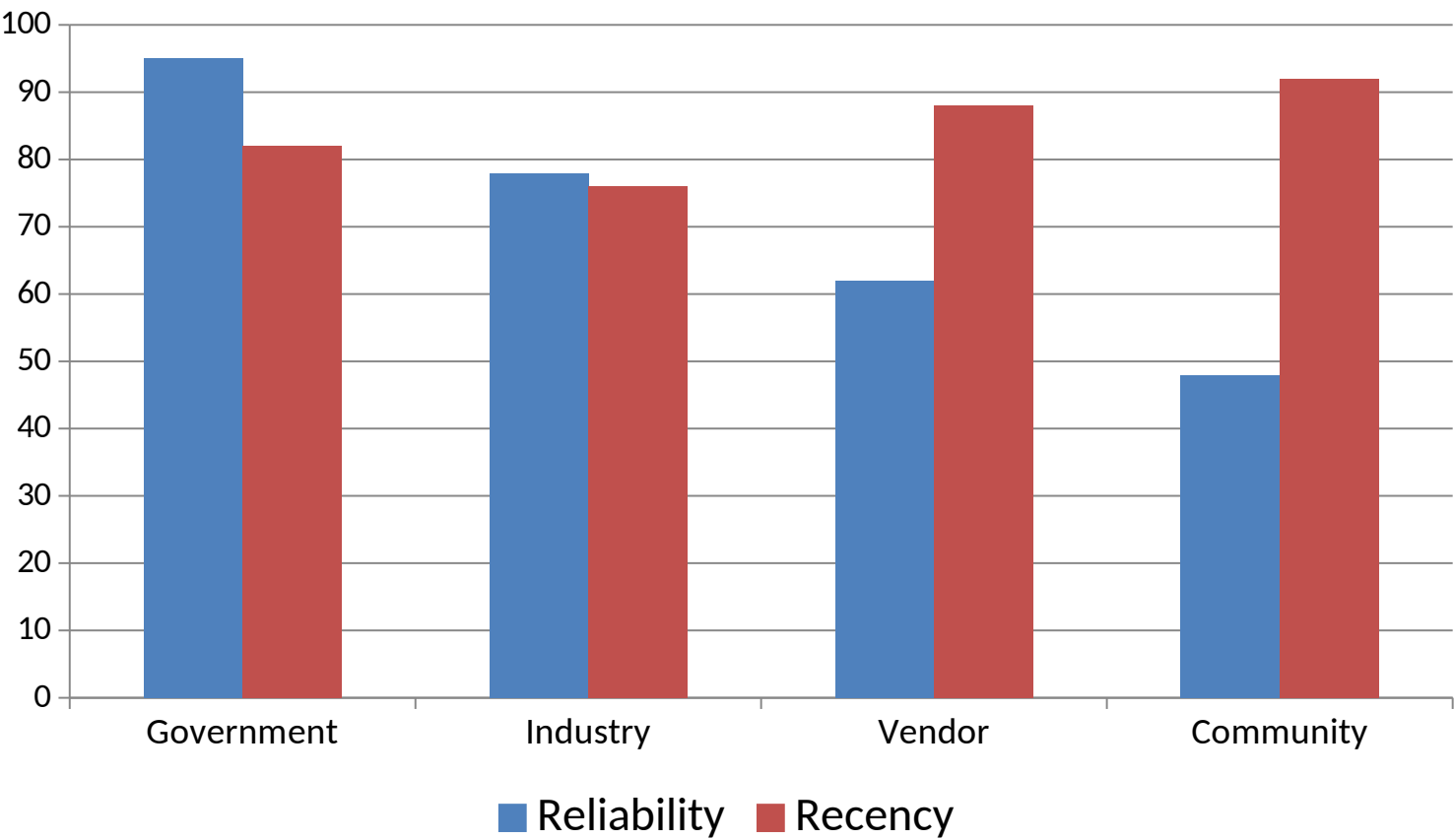
!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

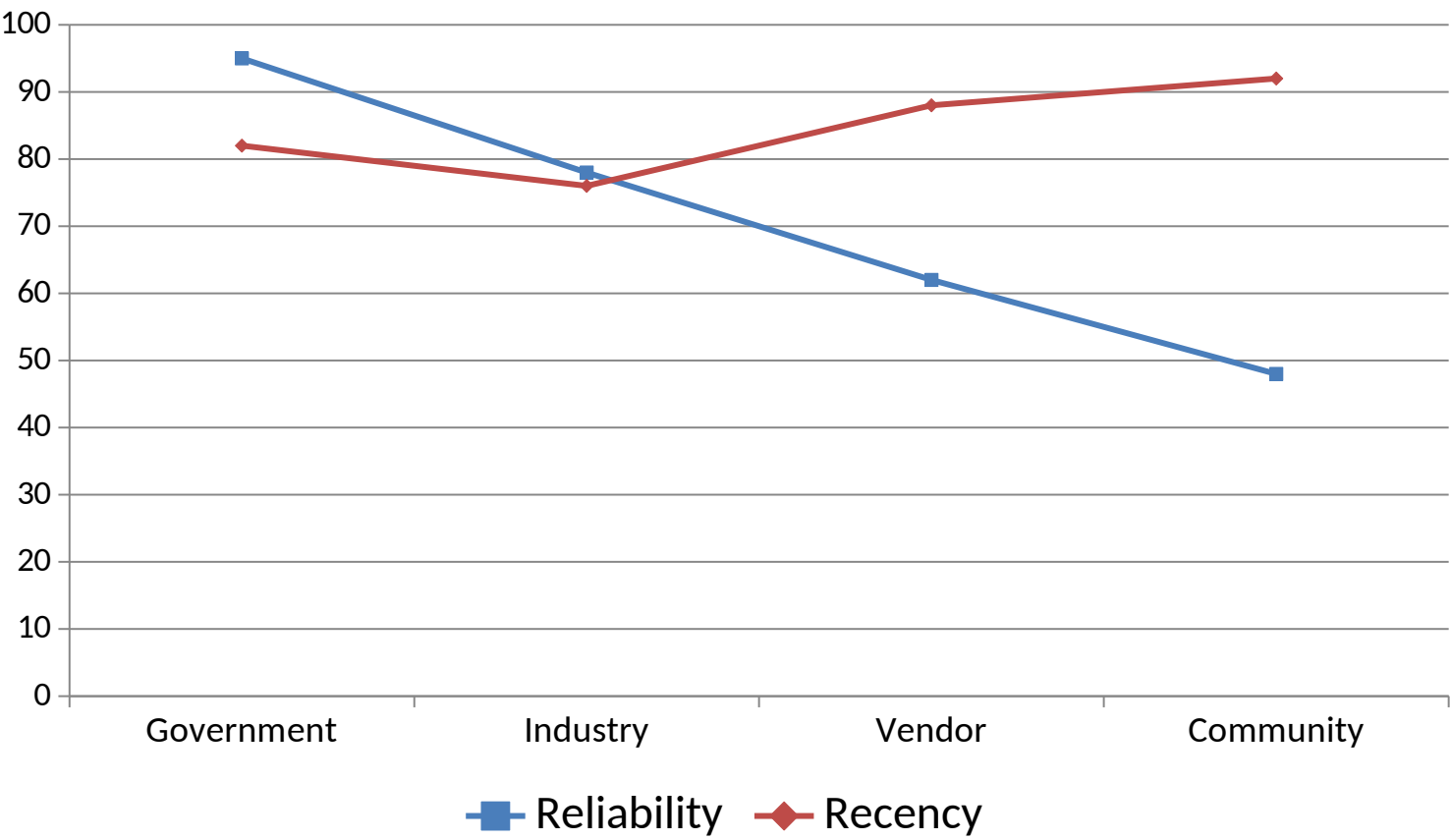
Topic 01 — worked dataset



What the data shows

Reliability and recency are separate; fresh does not always mean trustworthy.

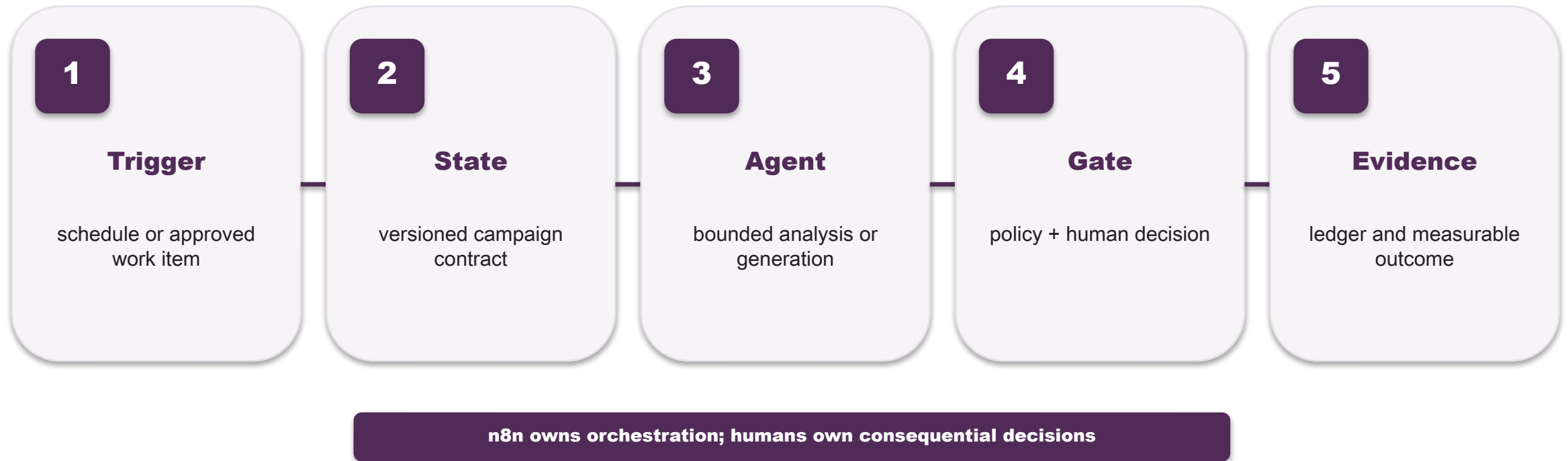
Topic 01 — trend and decision threshold



What the data shows

Read direction, volume and guardrails together; record what would change the decision.

Topic 01 — Control-Plane Architecture



Topic 01 — Control Case

1

Input conflict

Two plausible signals support different actions; one has stronger lineage and lower risk.

2

Decision rule

Prefer reproducible evidence, explicit assumptions and reversibility. Escalate unclear authority or risk.

3

Evidence

Record chosen input, rejected alternative, owner, threshold and next review date.

Lab 01 — Campaign Brief Intake & KPI Contract

S

Scenario

Build campaign stage 1 of the connected Northstar Launch marketing automation.

A

Artifact

A validated campaign object with objectives, KPI contracts, owners and guardrails

I

Input

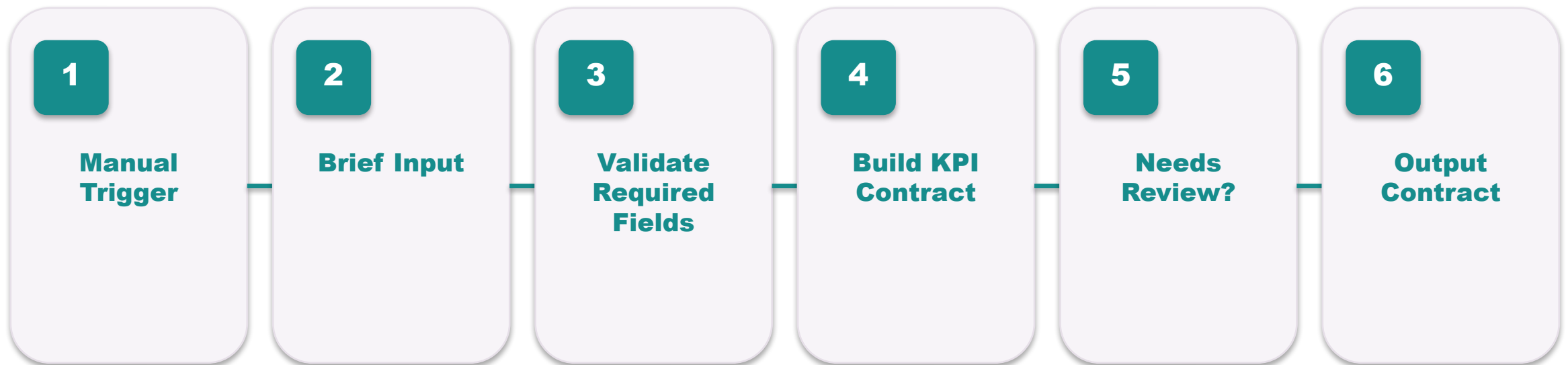
data/campaign_brief.xlsx · 8 named fields

✓

Acceptance

Output contains campaign_id CMP-NS-001, at least three KPI contracts, one owner per KPI and a human-review flag.

Lab 1 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 1 — input and output contract

FIELD 01	campaign_id	FIELD 02	objective
FIELD 03	audience	FIELD 04	budget_sgd
FIELD 05	start_date	FIELD 06	end_date
FIELD 07	owner	FIELD 08	risk_tier

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 1 — state and ownership



Lab 1 — metric and decision rule

$$\text{brief_completeness} = \text{populated_required_fields} / 8$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{brief_completeness} = \text{populated_required_fields} / 8$

Lab 1 — failure containment



Failure mode

Missing owner or budget can cause unbounded spend

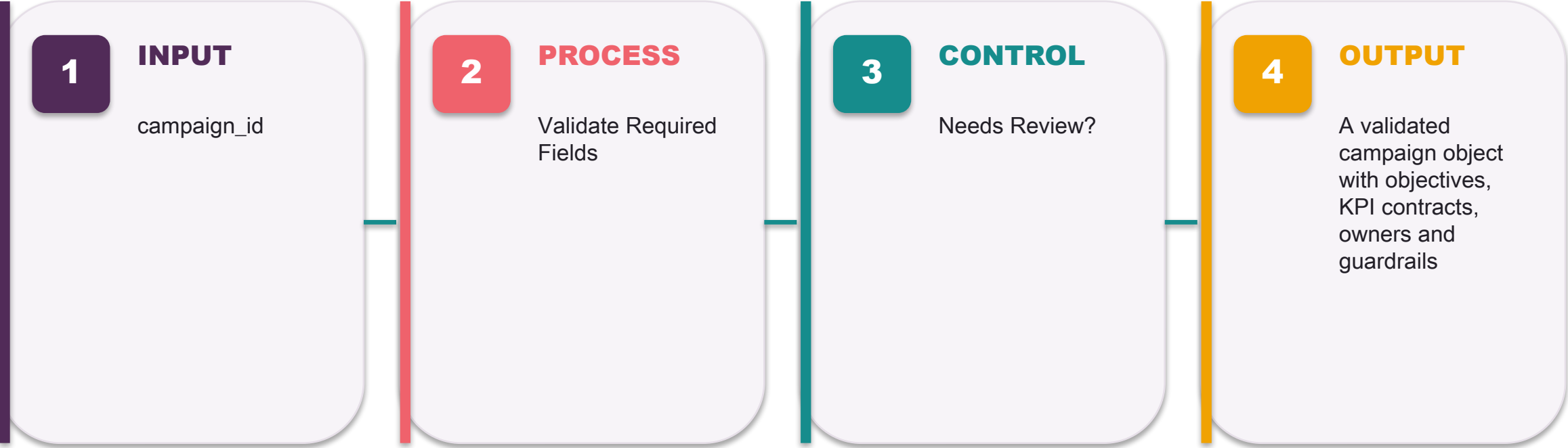


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 1 — Observable Evidence



Output contains campaign_id CMP-NS-001, at least three KPI contracts, one owner per KPI and a human-review flag.

Lab 1 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Missing owner or budget can cause unbounded spend

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: Output contains campaign_id CMP-NS-001, at least three KPI contracts, one owner per KPI and a human-review flag.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 1 — From Slide Model to Lab Evidence



Lab 02 — Evidence Research & Source Scoring

S

Scenario

Build campaign stage 2 of the connected Northstar Launch marketing automation.

A

Artifact

A research evidence table with claim, source, recency, reliability and composite score

I

Input

data/research_sources.xlsx · 8 named fields

✓

Acceptance

Every accepted claim retains source_id, URL, published date and evidence_score; no rejected row reaches synthesis.

Lab 2 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 2 — input and output contract

FIELD 01 source_id	FIELD 02 publisher
FIELD 03 url	FIELD 04 published_date
FIELD 05 claim	FIELD 06 relevance
FIELD 07 reliability	FIELD 08 recency

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 2 — state and ownership



Lab 2 — metric and decision rule

$$\text{evidence_score} = \text{relevance} \times \text{reliability} \times \text{recency}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{evidence_score} = \text{relevance} \times \text{reliability} \times \text{recency}$

Lab 2 — failure containment



Failure mode

Low-quality sources can contaminate every downstream asset



Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 2 — Observable Evidence



Every accepted claim retains source_id, URL, published date and evidence_score; no rejected row reaches synthesis.

Lab 2 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Low-quality sources can contaminate every downstream asset

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: Every accepted claim retains source_id, URL, published date and evidence_score; no rejected row reaches synthesis.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 2 — From Slide Model to Lab Evidence



Lab 03 — Audience Signal Synthesis & Opportunity Ranking

S

Scenario

Build campaign stage 3 of the connected Northstar Launch marketing automation.

A

Artifact

Ranked audience opportunities linked to evidence IDs

I

Input

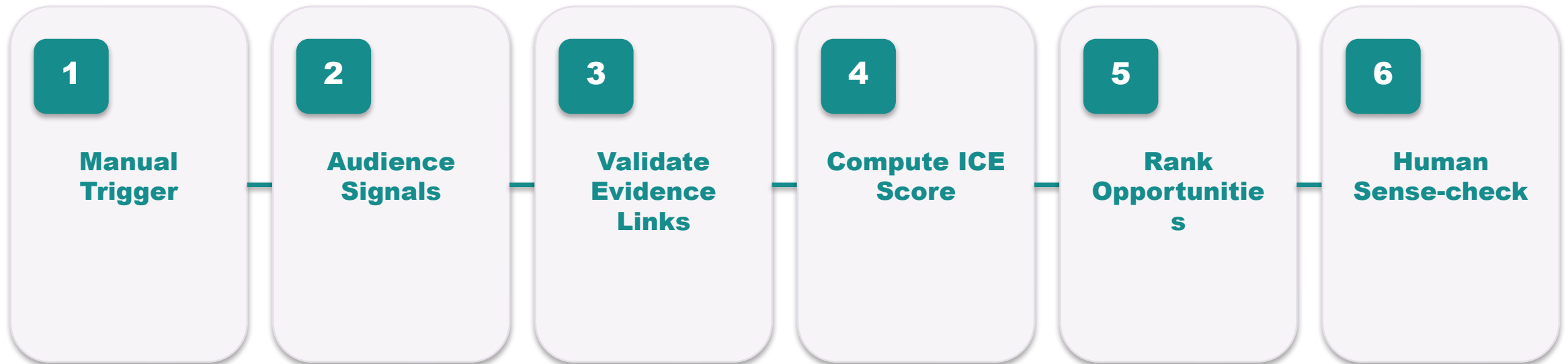
data/audience_signals.xlsx · 9 named fields

✓

Acceptance

Every opportunity has evidence_ids, an ICE score and explicit supported status before strategy synthesis.

Lab 3 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 3 — input and output contract

FIELD 01 signal_id	FIELD 02 segment
FIELD 03 job	FIELD 04 pain
FIELD 05 trigger	FIELD 06 evidence_ids
FIELD 07 impact	FIELD 08 confidence

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 3 — state and ownership



Lab 3 — metric and decision rule

$$\text{ICE} = \text{impact} \times \text{confidence} / \text{effort}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: ICE = impact × confidence / effort

Lab 3 — failure containment



Failure mode

AI-generated personas may encode stereotypes or unsupported claims

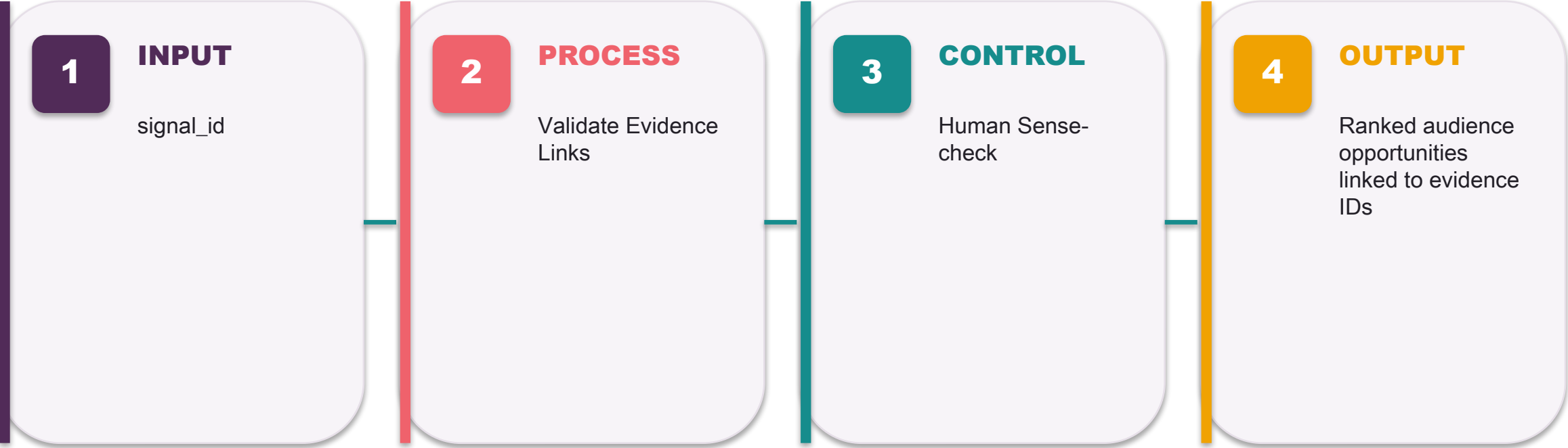


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 3 — Observable Evidence



Every opportunity has evidence_ids, an ICE score and explicit supported status before strategy synthesis.

Lab 3 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. AI-generated personas may encode stereotypes or unsupported claims

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: Every opportunity has evidence_ids, an ICE score and explicit supported status before strategy synthesis.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 3 — From Slide Model to Lab Evidence



TOPIC 02

Strategy, KPIs & ROI Models

RACE/SOSTAC · objectives · channel economics · budget decisions

LO1-LO2 · K1-K4 · A1-A4

Architecture · runtime · contract · controls · evidence



Objective tree

M

Mechanism

Business outcome →
marketing objective →
channel KPI → event and
owner.

C

Concrete contract

Named fields, owner,
timestamp, version and
acceptance status make this
mechanism testable.

!

Failure control

Reject missing lineage,
ambiguous ownership or
state transitions that exceed
delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

KPI contract

M

Mechanism

Name, formula, source, cadence, target, guardrail and action threshold.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Unit economics

M

Mechanism

CAC, LTV, contribution margin, ROAS and incremental ROI expose trade-offs.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Attribution limits

M

Mechanism

Last-click is observable but not causal; triangulate with experiments and holdouts.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Budget rules

M

Mechanism

Reallocate only when volume, confidence and guardrails exceed declared thresholds.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Strategy synthesis

M

Mechanism

RACE describes the journey;
SOSTAC governs planning
and control.

C

Concrete contract

Named fields, owner,
timestamp, version and
acceptance status make this
mechanism testable.

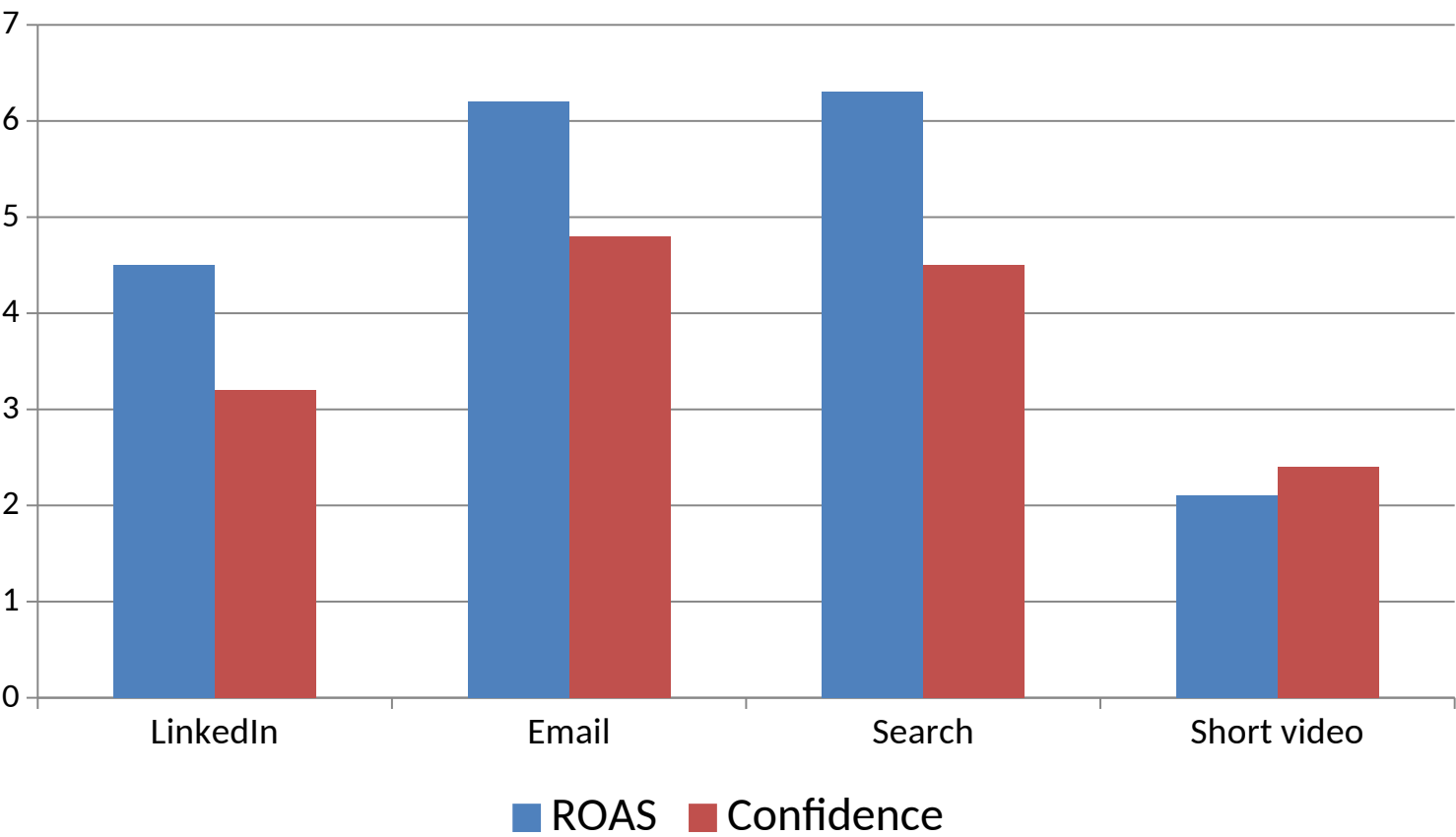
!

Failure control

Reject missing lineage,
ambiguous ownership or
state transitions that exceed
delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

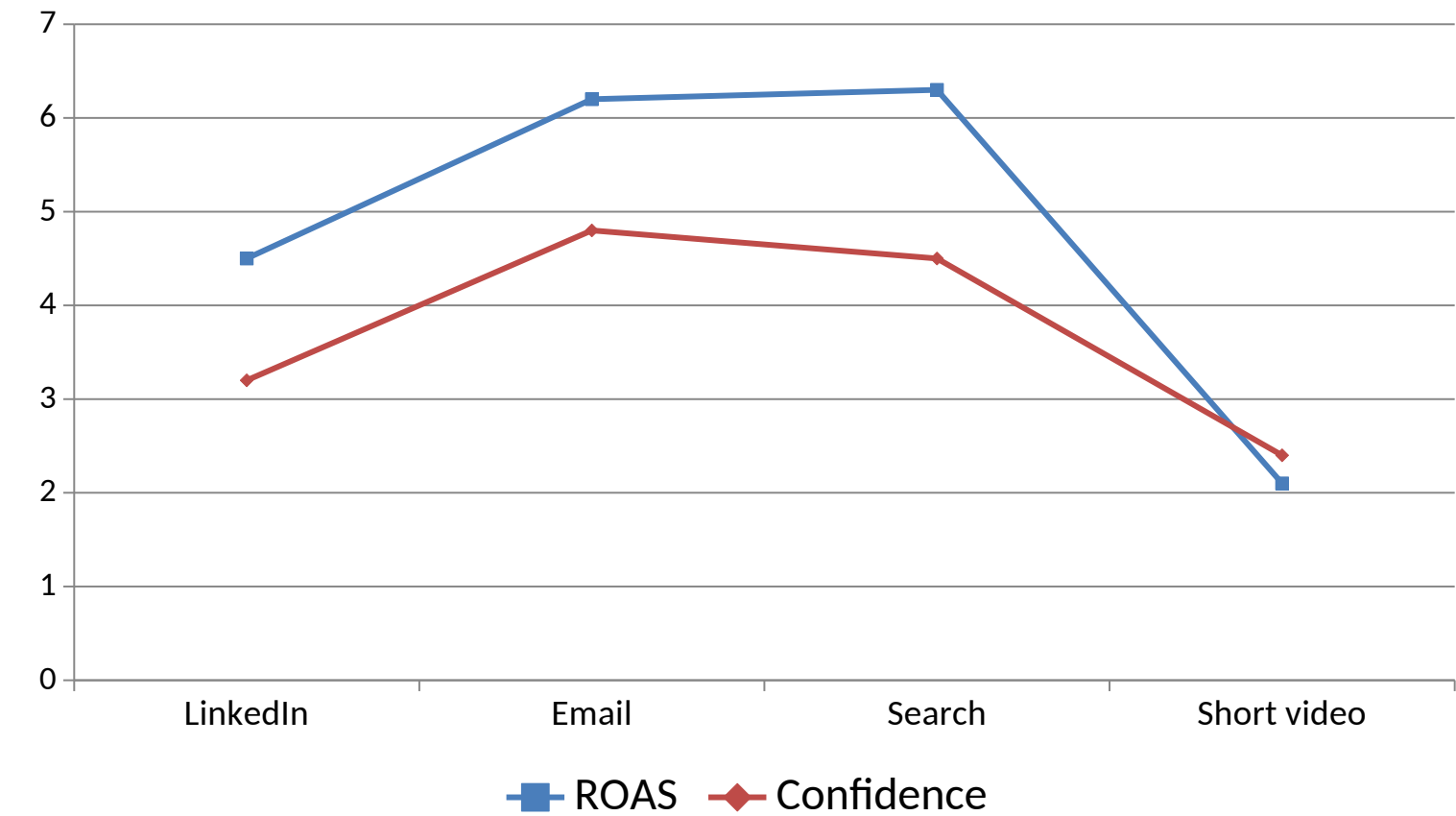
Topic 02 — worked dataset



What the data shows

Observed ROAS is not sufficient when confidence and assisted demand differ.

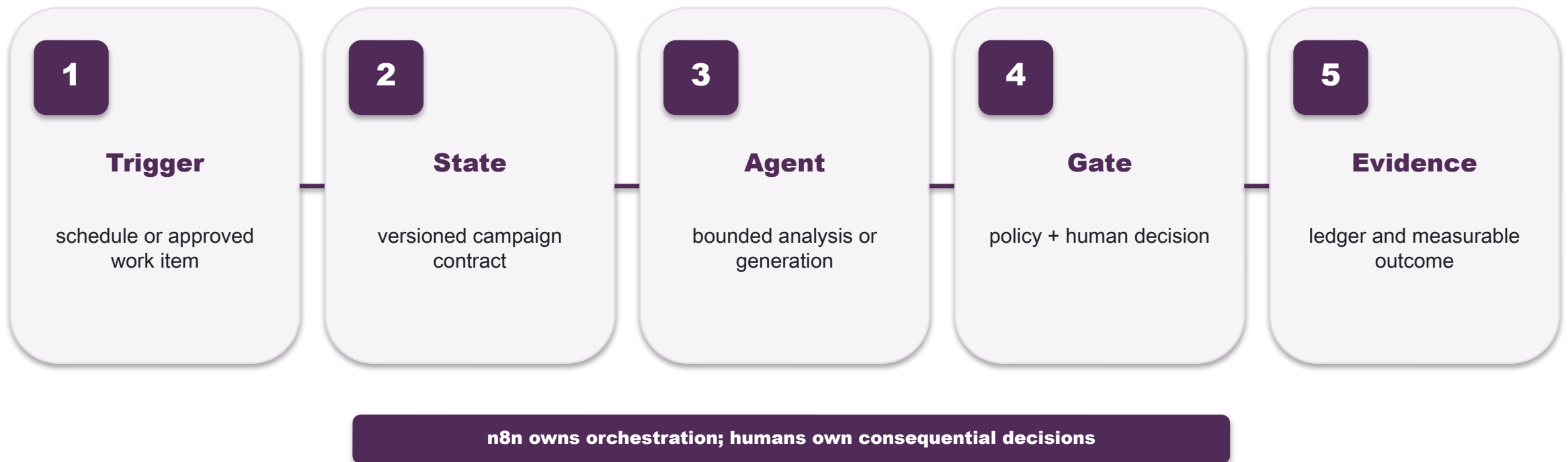
Topic 02 — trend and decision threshold



What the data shows

Read direction, volume and guardrails together; record what would change the decision.

Topic 02 — Control-Plane Architecture



Topic 02 — Control Case

1

Input conflict

Two plausible signals support different actions; one has stronger lineage and lower risk.

2

Decision rule

Prefer reproducible evidence, explicit assumptions and reversibility. Escalate unclear authority or risk.

3

Evidence

Record chosen input, rejected alternative, owner, threshold and next review date.

Lab 04 — RACE/SOSTAC Strategy Synthesizer

S

Scenario

Build campaign stage 4 of the connected Northstar Launch marketing automation.

A

Artifact

A strategy model connecting situation, objectives, audience, channels, tactics, actions and control

I

Input

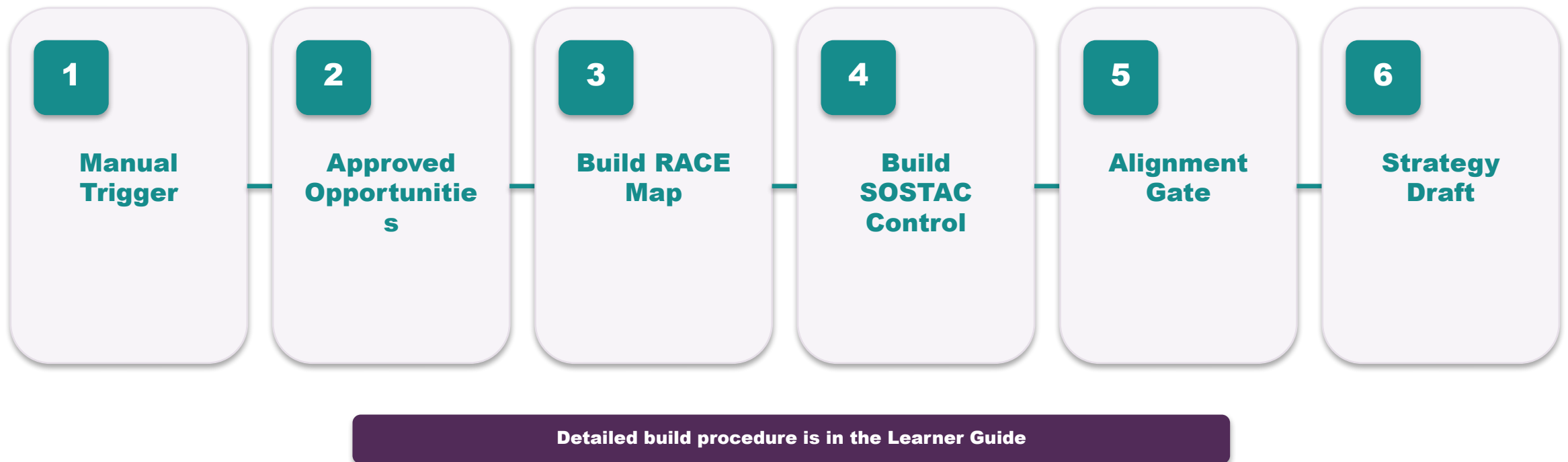
data/strategy_model.xlsx · 7 named fields

✓

Acceptance

All strategy decisions trace to evidence, map to RACE, include control fields and pass human strategy review.

Lab 4 — n8n execution path



Lab 4 — input and output contract

FIELD 01 stage	FIELD 02 decision
FIELD 03 evidence_id	FIELD 04 owner
FIELD 05 kpi	FIELD 06 target
FIELD 07 review_date	

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 4 — state and ownership



Lab 4 — metric and decision rule

$$\text{alignment_rate} = \text{decisions_with_objective_and_owner} / \text{total_decisions}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{alignment_rate} = \text{decisions_with_objective_and_owner} / \text{total_decisions}$

Lab 4 — failure containment



Failure mode

Channel tactics can drift from the business objective

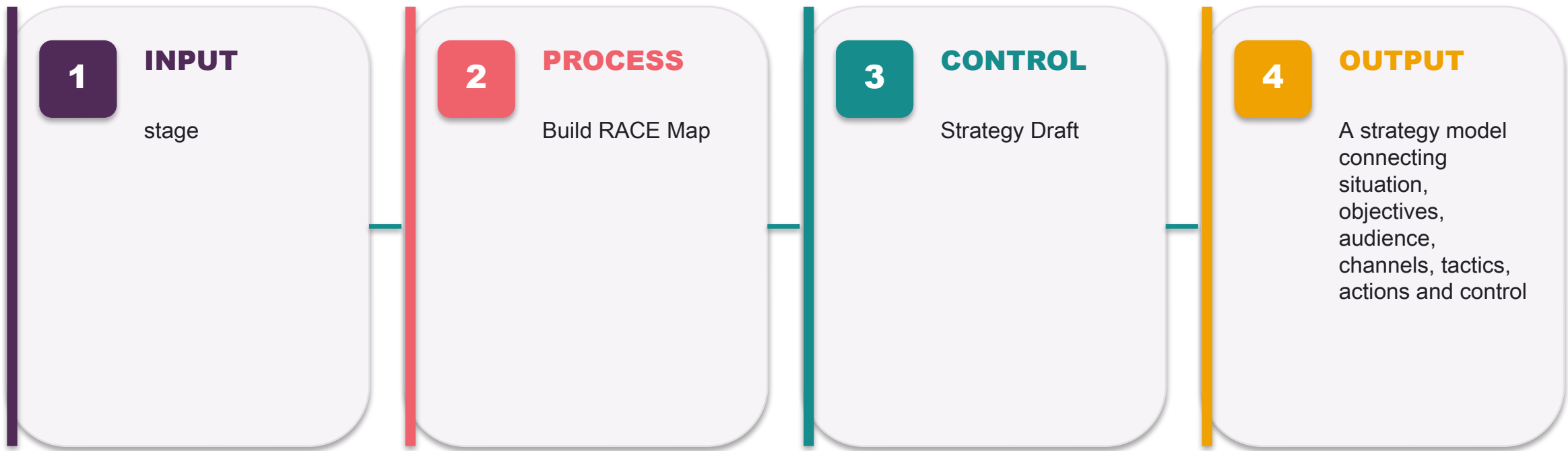


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 4 — Observable Evidence



All strategy decisions trace to evidence, map to RACE, include control fields and pass human strategy review.

Lab 4 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Channel tactics can drift from the business objective

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: All strategy decisions trace to evidence, map to RACE, include control fields and pass human strategy review.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 4 — From Slide Model to Lab Evidence



Lab 05 — Channel Budget & ROI Simulator

S

Scenario

Build campaign stage 5 of the connected Northstar Launch marketing automation.

A

Artifact

A scenario model for CAC, ROAS, contribution margin and incremental ROI

I

Input

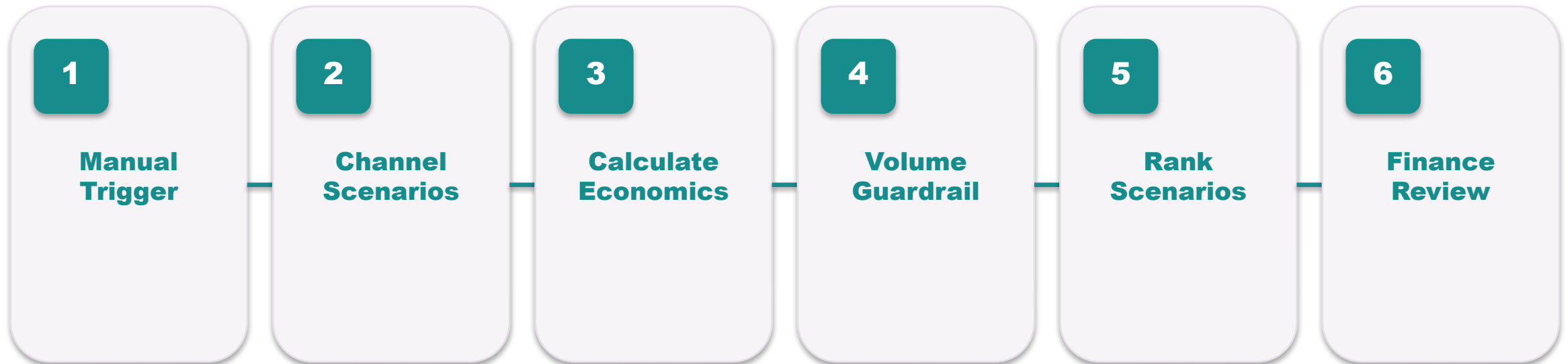
data/channel_economics.xlsx · 8 named fields

✓

Acceptance

Calculated metrics match Excel formulas, low-volume scenarios are flagged and no allocation proceeds without reviewer_decision.

Lab 5 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 5 — input and output contract

FIELD 01 channel	FIELD 02 spend
FIELD 03 impressions	FIELD 04 clicks
FIELD 05 leads	FIELD 06 customers
FIELD 07 revenue	FIELD 08 gross_margin

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 5 — state and ownership



Lab 5 — metric and decision rule

$\text{CAC} = \text{spend}/\text{customers}$; $\text{ROAS} = \text{revenue}/\text{spend}$; $\text{ROI} = (\text{revenue} \times \text{margin} - \text{spend})/\text{spend}$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{CAC} = \text{spend}/\text{customers}$; $\text{ROAS} = \text{revenue}/\text{spend}$; $\text{ROI} = (\text{revenue} \times \text{margin} - \text{spend})/\text{spend}$

Lab 5 — failure containment



Failure mode

Optimising ROAS alone may cut channels that create assisted demand

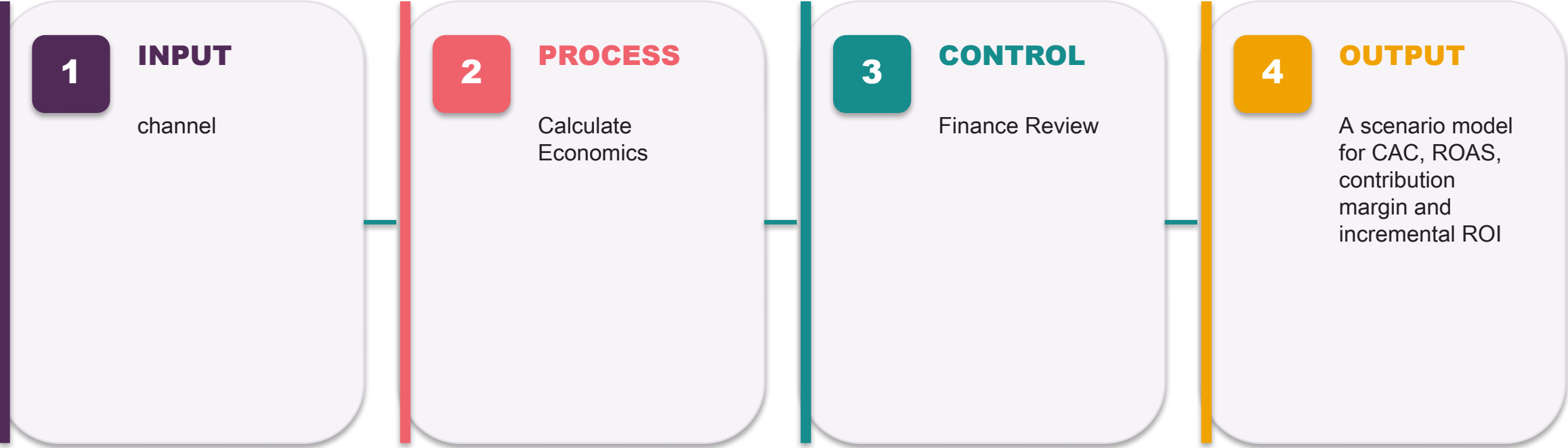


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 5 — Observable Evidence



Calculated metrics match Excel formulas, low-volume scenarios are flagged and no allocation proceeds without reviewer_decision.

Lab 5 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Optimising ROAS alone may cut channels that create assisted demand

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: Calculated metrics match Excel formulas, low-volume scenarios are flagged and no allocation proceeds without reviewer_decision.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 5 — From Slide Model to Lab Evidence



Lab 06 — Campaign Backlog & Experiment Design

S

Scenario

Build campaign stage 6 of the connected Northstar Launch marketing automation.

A

Artifact

A prioritised backlog with hypotheses, variants, sample thresholds and stop rules

I

Input

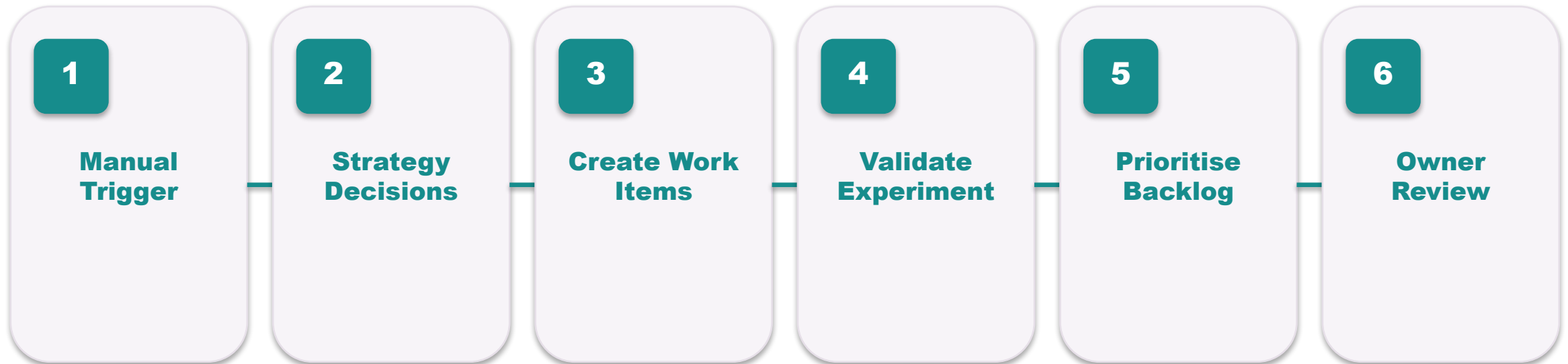
data/campaign_backlog.xlsx · 8 named fields

✓

Acceptance

Each ready item has one falsifiable hypothesis, primary metric, minimum sample, stop rule and accountable owner.

Lab 6 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 6 — input and output contract

FIELD 01 item_id	FIELD 02 stage
FIELD 03 hypothesis	FIELD 04 primary_metric
FIELD 05 minimum_sample	FIELD 06 stop_rule
FIELD 07 owner	FIELD 08 status

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 6 — state and ownership



Lab 6 — metric and decision rule

$$\text{priority} = \text{expected_impact} \times \text{confidence} / \text{effort}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{priority} = \text{expected_impact} \times \text{confidence} / \text{effort}$

Lab 6 — failure containment



Failure mode

Automated testing without stop rules can expose audiences to harmful variants

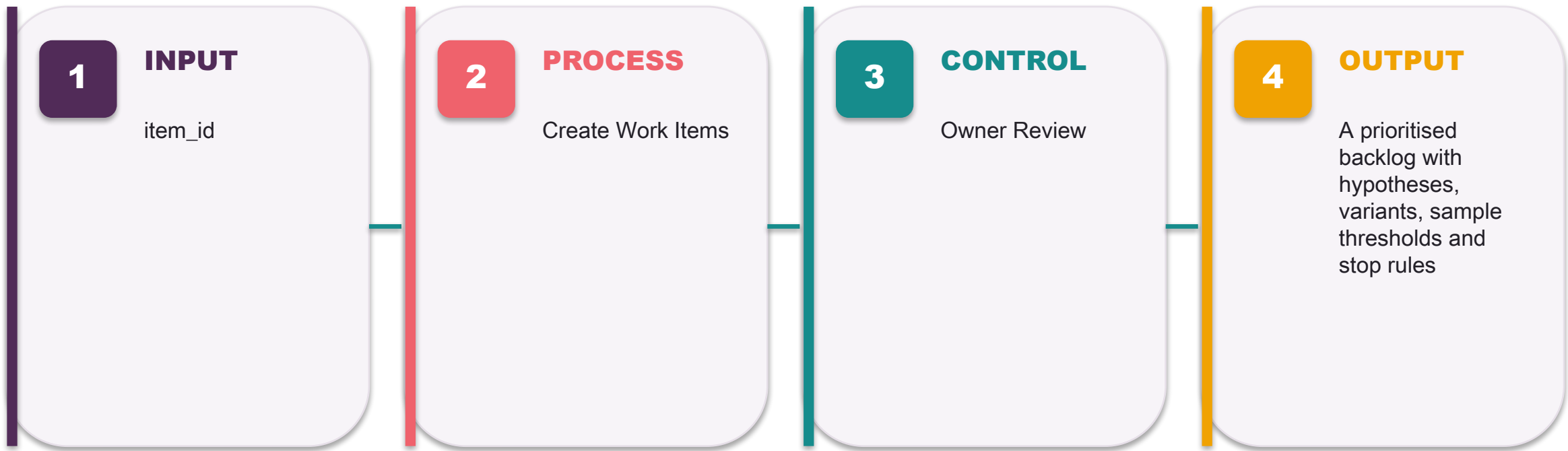


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 6 — Observable Evidence



Each ready item has one falsifiable hypothesis, primary metric, minimum sample, stop rule and accountable owner.

Lab 6 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Automated testing without stop rules can expose audiences to harmful variants

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: Each ready item has one falsifiable hypothesis, primary metric, minimum sample, stop rule and accountable owner.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 6 — From Slide Model to Lab Evidence



TOPIC 03

Agentic Content Operations

Brief contracts · generation · brand controls · reusable channel transformations

LO3 · K3/K5 · A4–A5

Architecture · runtime · contract · controls · evidence



Content contract

M

Mechanism

One canonical brief becomes channel-specific outputs through explicit schemas.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Prompt boundary

M

Mechanism

System rules, evidence context, task, format and rejection conditions are separated.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Brand retrieval

M

Mechanism

Approved claims, tone rules and prohibited language are fetched before generation.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Channel transform

M

Mechanism

The same message is adapted to intent, format, length and CTA—not copied verbatim.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Quality gate

M

Mechanism

Schema, claim, tone, privacy and accessibility checks run before human review.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Idempotency

M

Mechanism

A campaign-content key prevents duplicate generation when workflows retry.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

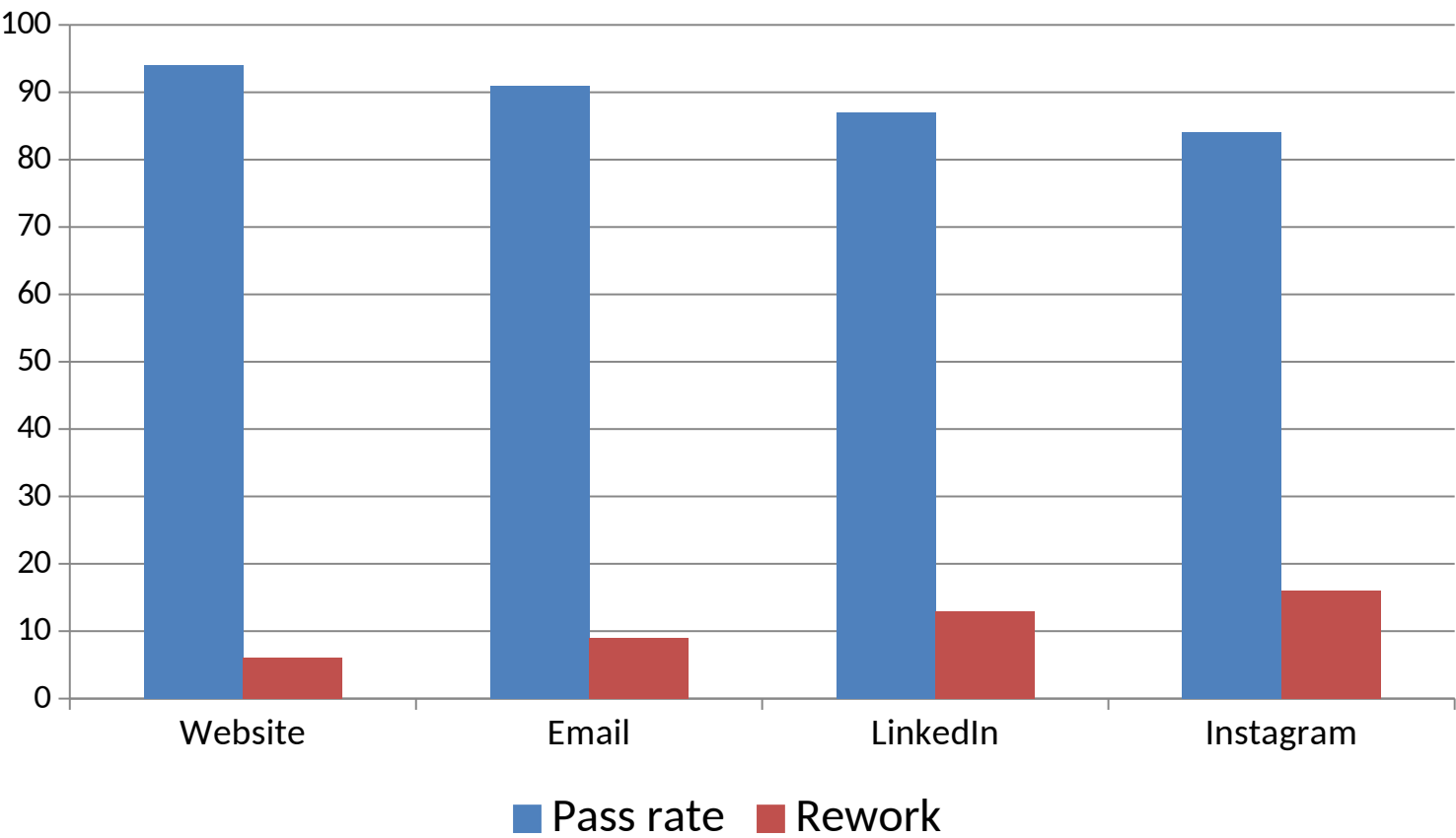
!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

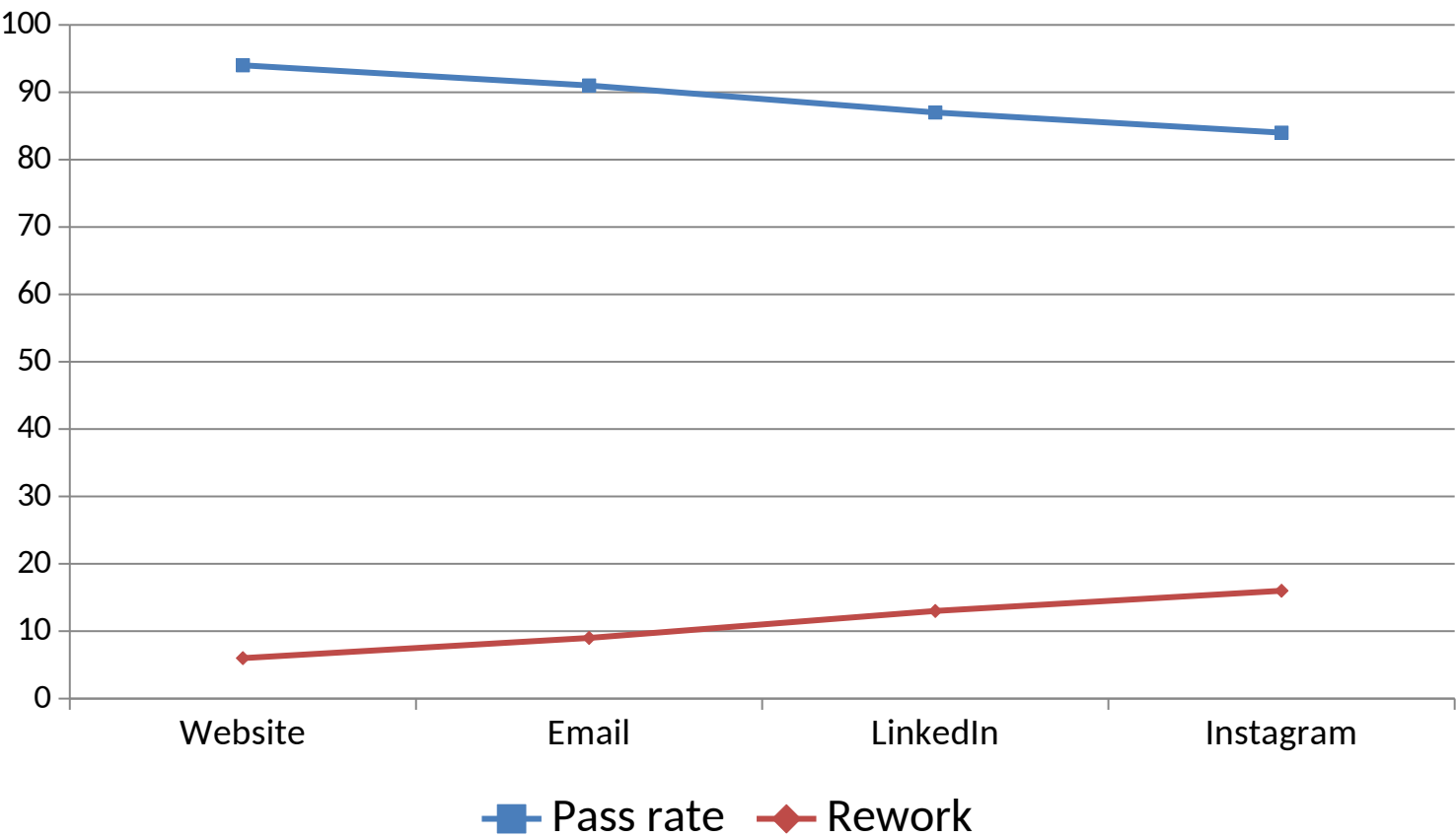
Topic 03 — worked dataset



What the data shows

Channel-specific transforms reduce rework.

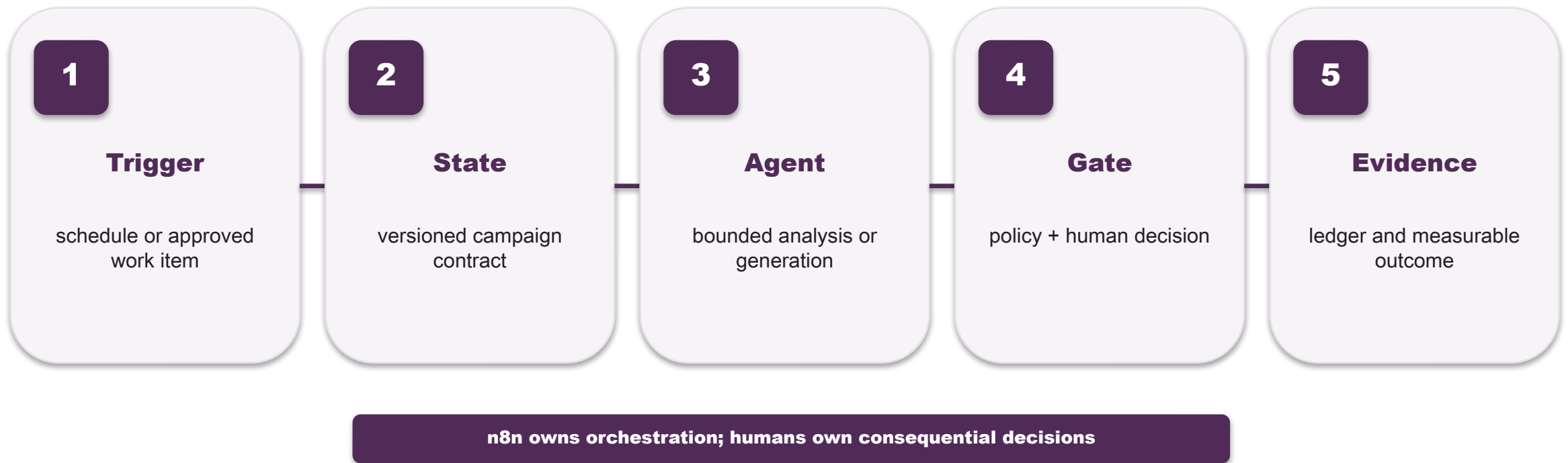
Topic 03 — trend and decision threshold



What the data shows

Read direction, volume and guardrails together; record what would change the decision.

Topic 03 — Control-Plane Architecture



Topic 03 — Control Case

1

Input conflict

Two plausible signals support different actions; one has stronger lineage and lower risk.

2

Decision rule

Prefer reproducible evidence, explicit assumptions and reversibility. Escalate unclear authority or risk.

3

Evidence

Record chosen input, rejected alternative, owner, threshold and next review date.

Lab 07 — Content Brief & Prompt Contract

S

Scenario

Build campaign stage 7 of the connected Northstar Launch marketing automation.

A

Artifact

A versioned JSON content contract suitable for deterministic channel generation

I

Input

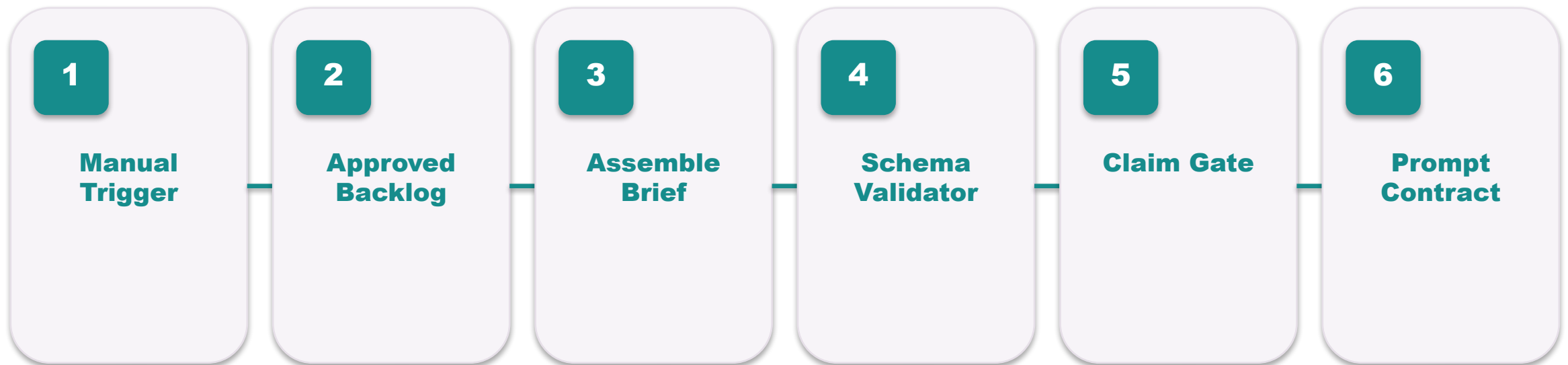
data/content_brief.xlsx · 9 named fields

✓

Acceptance

The approved contract includes evidence IDs, approved claims, prohibited terms, CTA, version and contract hash.

Lab 7 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 7 — input and output contract

FIELD 01 brief_id	FIELD 02 campaign_id
FIELD 03 audience_job	FIELD 04 message
FIELD 05 approved_claims	FIELD 06 evidence_ids
FIELD 07 tone	FIELD 08 cta

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 7 — state and ownership



Lab 7 — metric and decision rule

**contract_valid = required_fields_present AND
evidence_ids_resolved**



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: contract_valid = required_fields_present AND evidence_ids_resolved

Lab 7 — failure containment



Failure mode

Unbounded prompts can invent claims or leak confidential context

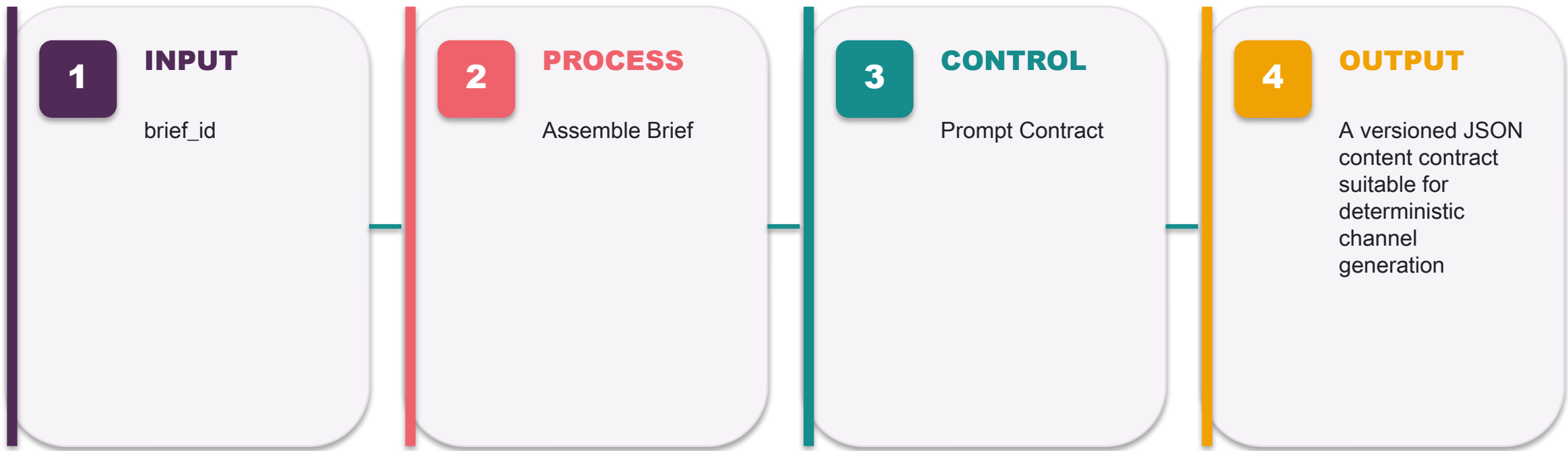


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 7 — Observable Evidence



The approved contract includes evidence IDs, approved claims, prohibited terms, CTA, version and contract hash.

Lab 7 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Unbounded prompts can invent claims or leak confidential context

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets:
The approved contract includes evidence IDs, approved claims, prohibited terms, CTA, version and contract hash.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 7 — From Slide Model to Lab Evidence



Lab 08 — Multi-Channel Content Factory

S

Scenario

Build campaign stage 8 of the connected Northstar Launch marketing automation.

A

Artifact

Website, email and SocialPost text/photo/video drafts with shared message lineage

I

Input

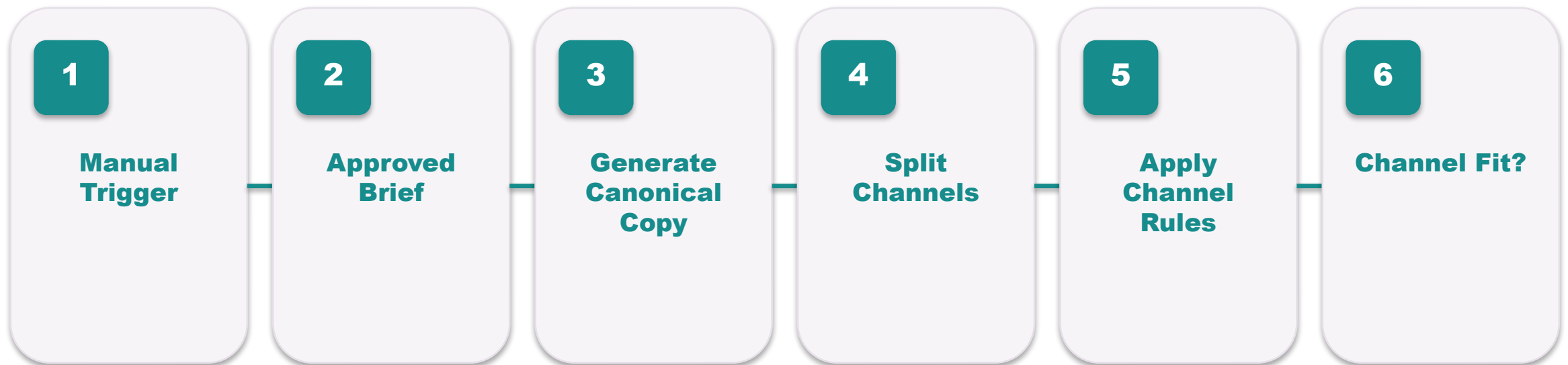
data/content_matrix.xlsx · 14 named fields

✓

Acceptance

One approved brief produces distinct channel assets; every social asset is endpoint-ready and retains brief_id, evidence_ids and validation status.

Lab 8 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 8 — input and output contract

FIELD 01 asset_id	FIELD 02 brief_id
FIELD 03 evidence_ids	FIELD 04 channel
FIELD 05 media_type	FIELD 06 title
FIELD 07 description	FIELD 08 caption

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 8 — state and ownership



Lab 8 — metric and decision rule

channel_fit = passed_length AND required_fields AND approved_claims_only



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: channel_fit = passed_length AND required_fields AND approved_claims_only

Lab 8 — failure containment



Failure mode

Copying identical text across channels ignores audience intent and platform constraints

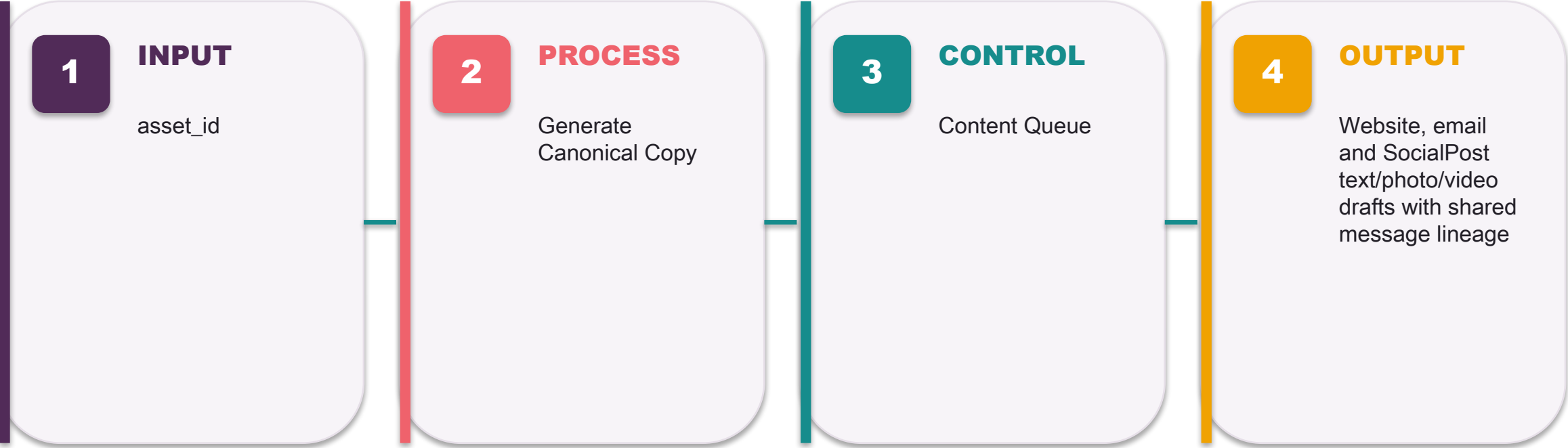


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 8 — Observable Evidence



One approved brief produces distinct channel assets; every social asset is endpoint-ready and retains brief_id, evidence_ids and validation status.

Lab 8 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Copying identical text across channels ignores audience intent and platform constraints

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: One approved brief produces distinct channel assets; every social asset is endpoint-ready and retains `brief_id`, `evidence_ids` and `validation status`.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 8 — From Slide Model to Lab Evidence



Lab 09 — Brand, Claim & Compliance QA

S

Scenario

Build campaign stage 9 of the connected Northstar Launch marketing automation.

A

Artifact

A QA report covering schema, brand, claims, privacy, accessibility and channel limits

I

Input

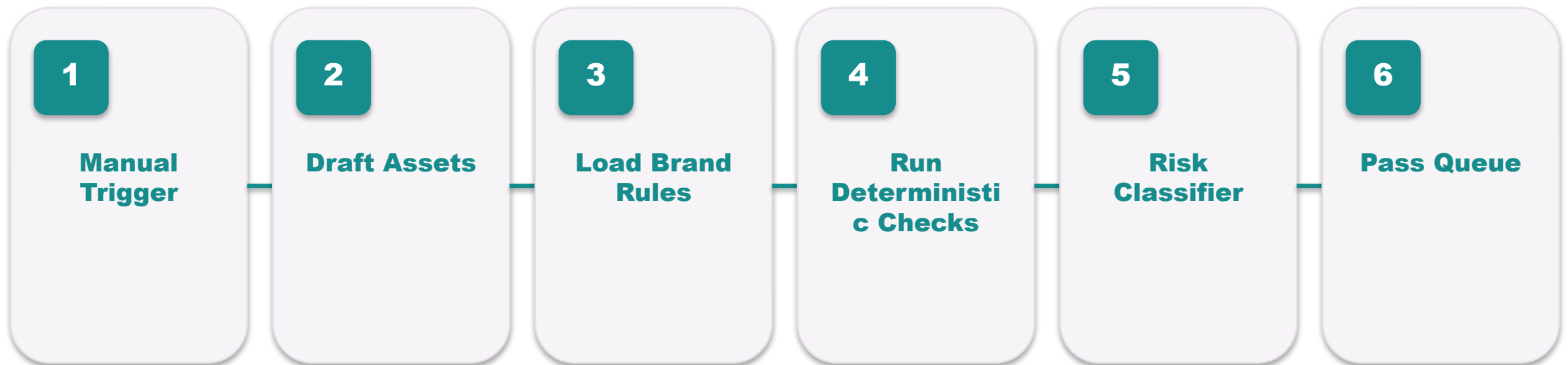
data/brand_rules.xlsx · 6 named fields

✓

Acceptance

No high-severity asset reaches approval; every medium-risk finding has category, evidence, owner and disposition.

Lab 9 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 9 — input and output contract

FIELD 01 rule_id	FIELD 02 category
FIELD 03 pattern	FIELD 04 severity
FIELD 05 action	FIELD 06 owner

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 9 — state and ownership



Lab 9 — metric and decision rule

$$\text{qa_pass_rate} = \text{passed_rules} / \text{applicable_rules}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{qa_pass_rate} = \text{passed_rules} / \text{applicable_rules}$

Lab 9 — failure containment



Failure mode

An AI self-review can miss the same hallucination made during generation

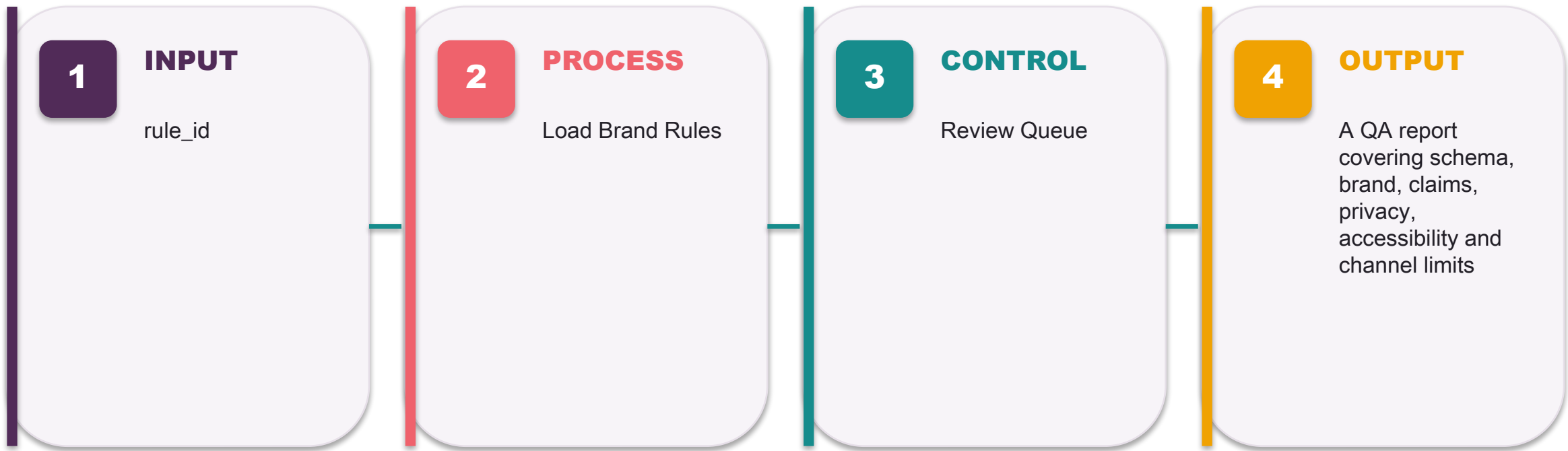


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 9 — Observable Evidence



No high-severity asset reaches approval; every medium-risk finding has category, evidence, owner and disposition.

Lab 9 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. An AI self-review can miss the same hallucination made during generation

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets:
No high-severity asset reaches approval; every medium-risk finding has category, evidence, owner and disposition.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 9 — From Slide Model to Lab Evidence



TOPIC 04

Human Oversight & Omnichannel Publishing

Approval states · SocialPost API · scheduling · idempotency · audit trail

LO3 · K5 · A5

Architecture · runtime · contract · controls · evidence



State machine

M

Mechanism

DRAFT → REVIEW →
APPROVED/REJECTED →
SCHEDULED → PUBLISHED
→ MEASURED.

C

Concrete contract

Named fields, owner,
timestamp, version and
acceptance status make this
mechanism testable.

!

Failure control

Reject missing lineage,
ambiguous ownership or
state transitions that exceed
delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Human-in-the-loop

M

Mechanism

High-impact tool calls pause until an accountable reviewer approves the exact payload.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Separation of duties

M

Mechanism

The authoring agent cannot self-approve or silently expand channel scope.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

SocialPost contract

M

Mechanism

n8n selects /api/upload, /api/upload_photos or /api/upload_text, then sends approved multipart fields with an Apikey credential.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Recovery

M

Mechanism

Retry only transient errors; dead-letter permanent failures with evidence and owner.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Auditability

M

Mechanism

Store reviewer, decision, timestamp, version, payload hash and publication response.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

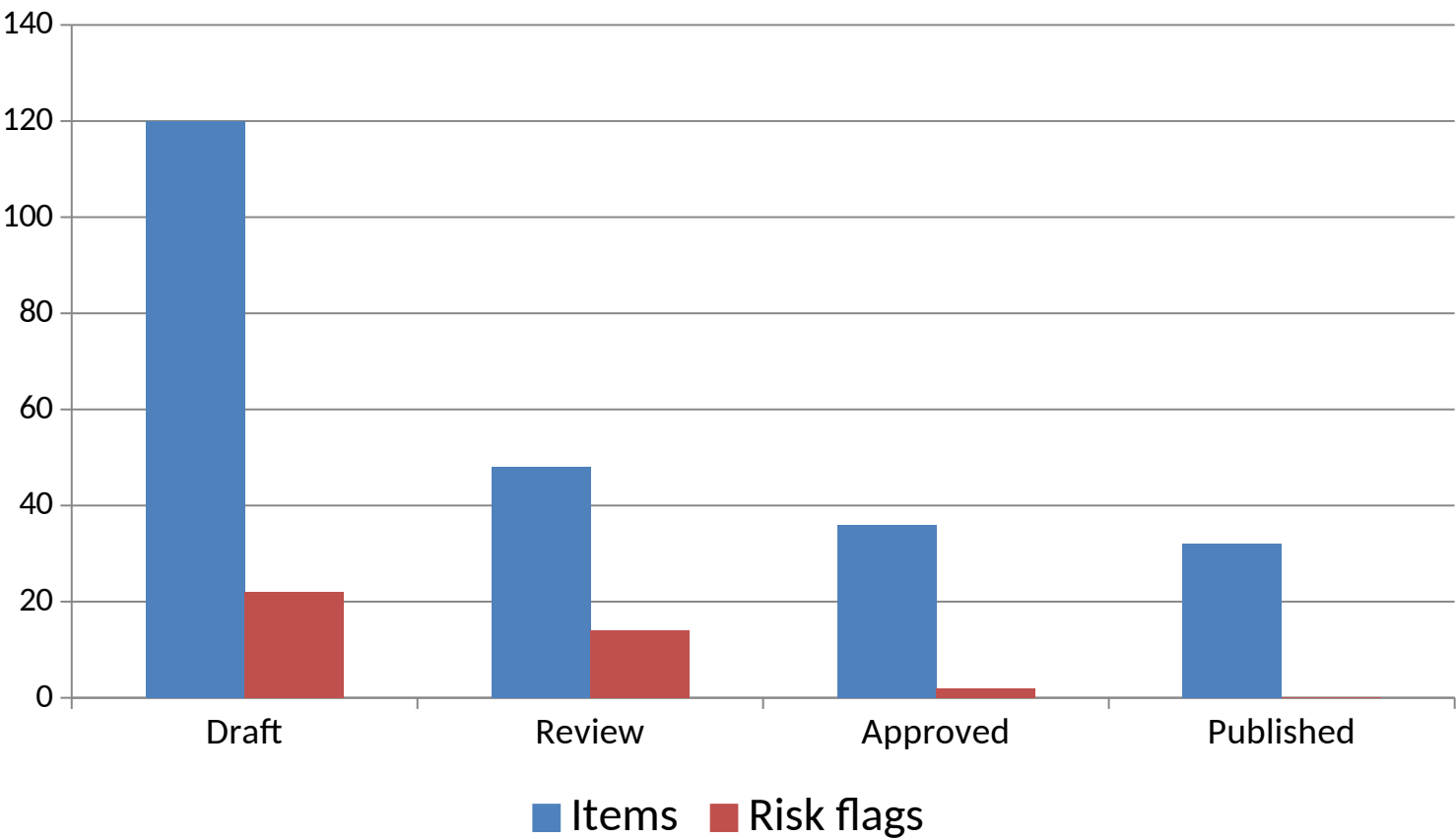
!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

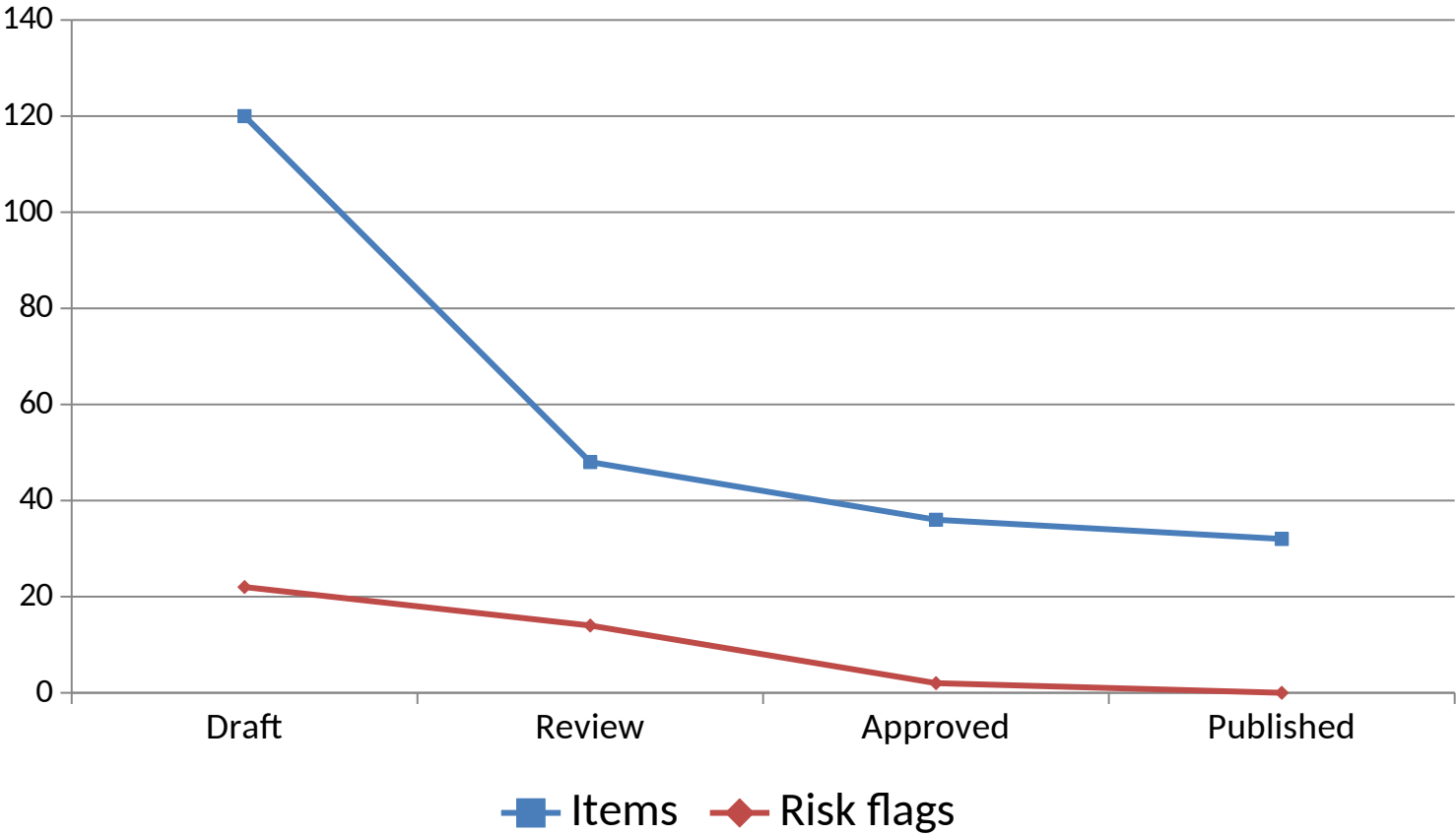
Topic 04 — worked dataset



What the data shows

The funnel removes or remediates risk before publication.

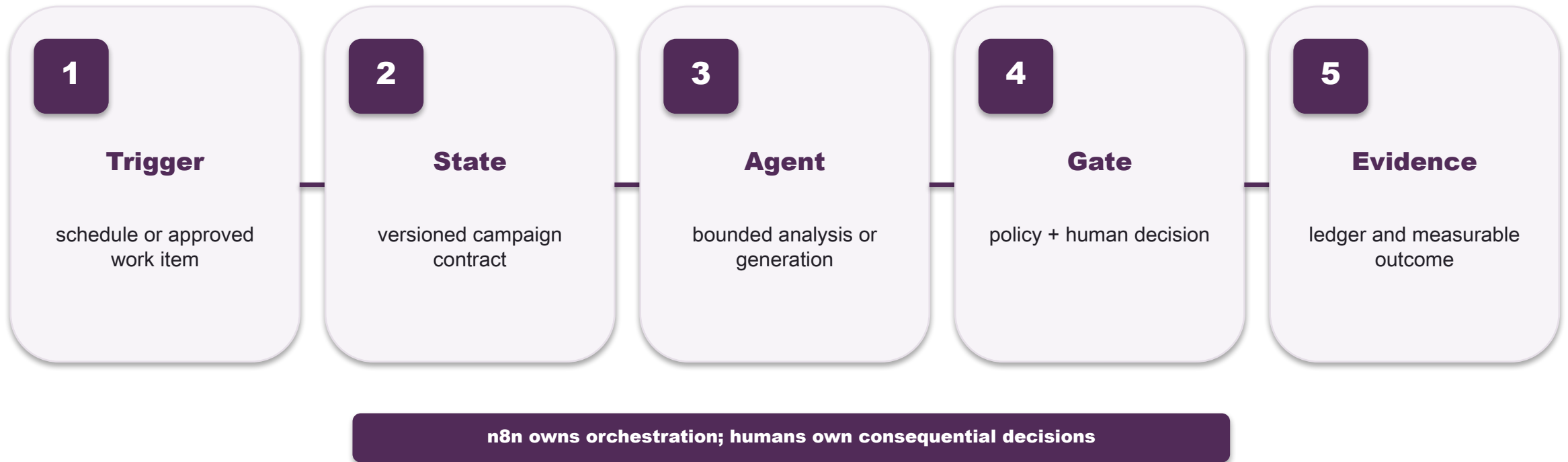
Topic 04 — trend and decision threshold



What the data shows

Read direction, volume and guardrails together; record what would change the decision.

Topic 04 — Control-Plane Architecture



Topic 04 — Control Case

1

Input conflict

Two plausible signals support different actions; one has stronger lineage and lower risk.

2

Decision rule

Prefer reproducible evidence, explicit assumptions and reversibility. Escalate unclear authority or risk.

3

Evidence

Record chosen input, rejected alternative, owner, threshold and next review date.

SocialPost API — Endpoint Selection



TEXT POST

POST /api/upload_text
user · platform[] · title · caption



PHOTO ALBUM

POST /api/upload_photos
photos[] · user · platform[] · title
· description



VIDEO

POST /api/upload
video · title · user · platform[]

n8n chooses the endpoint from media_type; it never changes media type after approval.

n8n → SocialPost — Multipart Request

```
POST
https://socialmediapost.tertiaryinfotech.com/api/upload_text
```

```
Authorization: Apikey <n8n Header Auth credential>
Content-Type: multipart/form-data
```

```
user      = {{$json.user}}
platform[] = {{$json.platform}}
title     = {{$json.title}}
caption   = {{$json.caption}}
```

1

Credential boundary

Store the full Apikey header value only in n8n Credentials.

2

Approval boundary

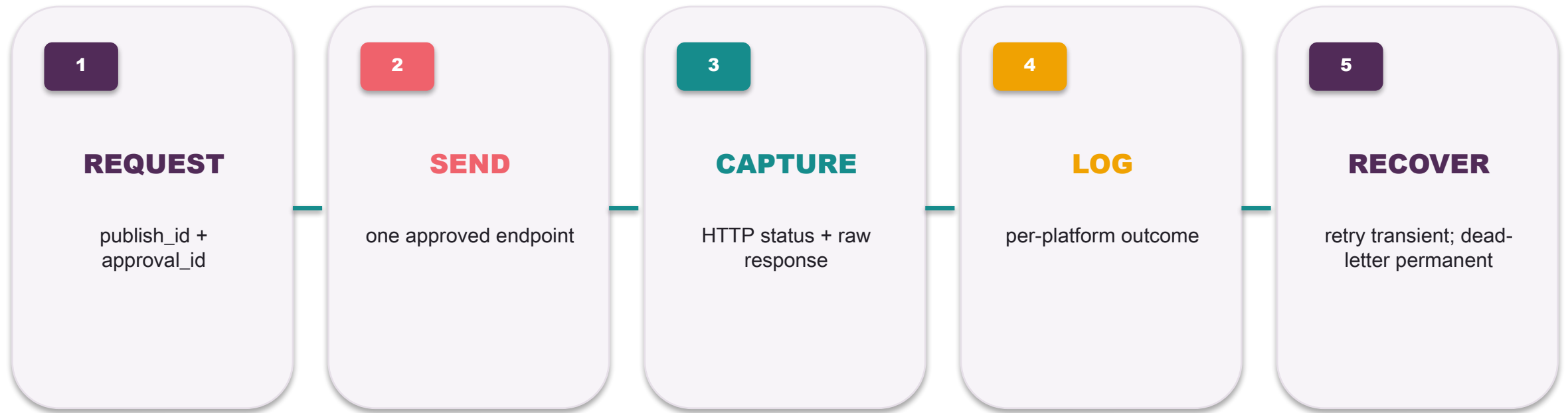
Hash endpoint, user, platform, title and caption before review.

3

Execution boundary

The supplied live node is disabled; dry-run inspection sends nothing.

SocialPost — Evidence and Recovery



Do not invent response fields: retain the raw response first, then map only observed keys.

Lab 10 — Human Approval State Machine

S

Scenario

Build campaign stage 10 of the connected Northstar Launch marketing automation.

A

Artifact

An approval request, tamper-evident decision record and state transition bound to endpoint, user and platforms

I

Input

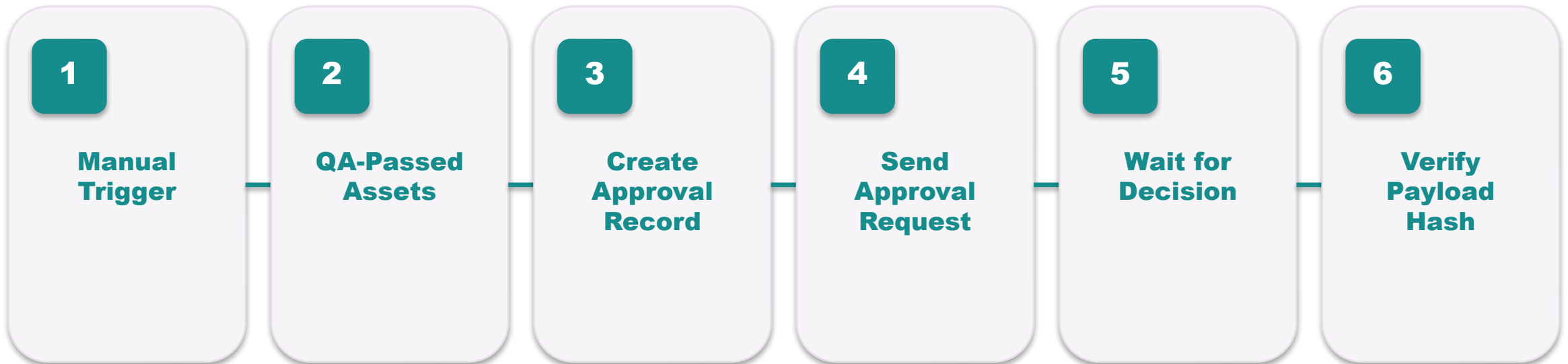
data/approval_queue.xlsx · 13 named fields

✓

Acceptance

Only APPROVED records with a matching SocialPost payload hash proceed; rejected, retargeted or changed payloads cannot publish.

Lab 10 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 10 — input and output contract

FIELD 01 approval_id	FIELD 02 asset_id
FIELD 03 api_path	FIELD 04 user
FIELD 05 platforms	FIELD 06 title
FIELD 07 caption	FIELD 08 decision_hash

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 10 — state and ownership



Lab 10 — metric and decision rule

**approval_integrity = decision_hash ==
current_payload_hash**



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: approval_integrity = decision_hash == current_payload_hash

Lab 10 — failure containment



Failure mode

A reviewer may approve one version while a changed payload is later published

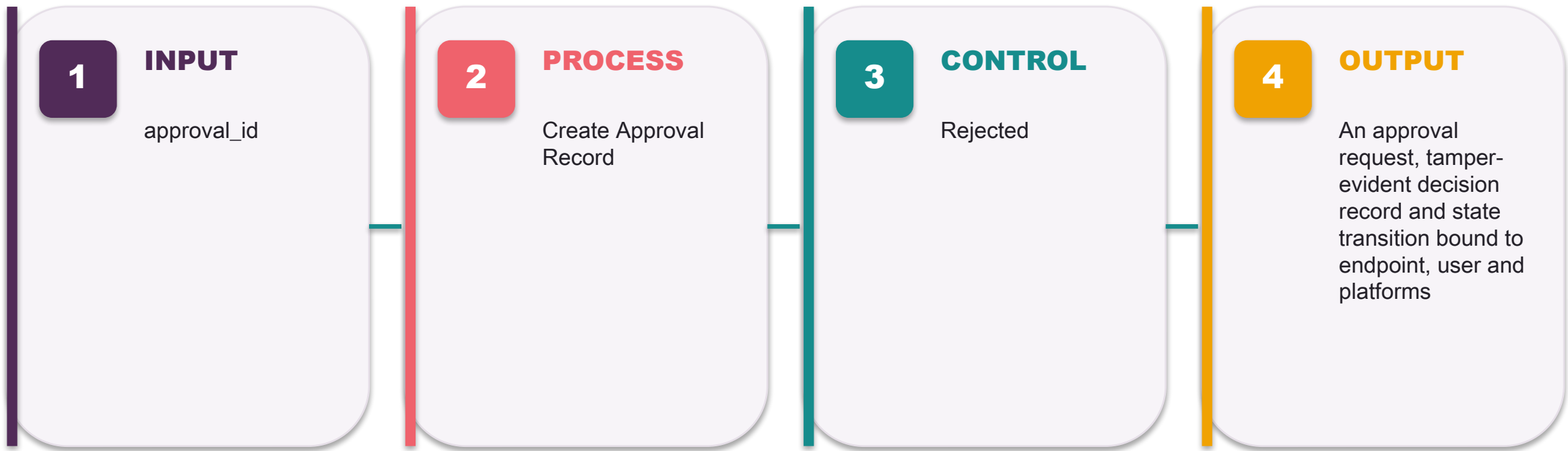


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 10 — Observable Evidence



Only APPROVED records with a matching SocialPost payload hash proceed; rejected, retargeted or changed payloads cannot publish.

Lab 10 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. A reviewer may approve one version while a changed payload is later published

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: Only APPROVED records with a matching SocialPost payload hash proceed; rejected, retargeted or changed payloads cannot publish.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 10 — From Slide Model to Lab Evidence



Lab 11 — SocialPost API Publishing & Idempotency

S

Scenario

Build campaign stage 11 of the connected Northstar Launch marketing automation.

A

Artifact

Validated SocialPost multipart request; live HTTP node disabled

I

Input

data/publishing_queue.xlsx · 18 named fields

✓

Acceptance

The inspected request uses the documented SocialPost endpoint and Apikey credential pattern, remains approval-linked and idempotent, and sends nothing unless the trainer explicitly enables the sandbox node.

Lab 11 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 11 — input and output contract

FIELD 01 publish_id	FIELD 02 asset_id
FIELD 03 media_type	FIELD 04 api_path
FIELD 05 user	FIELD 06 platforms
FIELD 07 title	FIELD 08 description

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 11 — state and ownership



Lab 11 — metric and decision rule

$$\text{duplicate_rate} = \text{duplicate_attempts} / \text{publish_attempts}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{duplicate_rate} = \text{duplicate_attempts} / \text{publish_attempts}$

Lab 11 — failure containment



Failure mode

Retries can publish the same approved content more than once or to an unapproved SocialPost profile

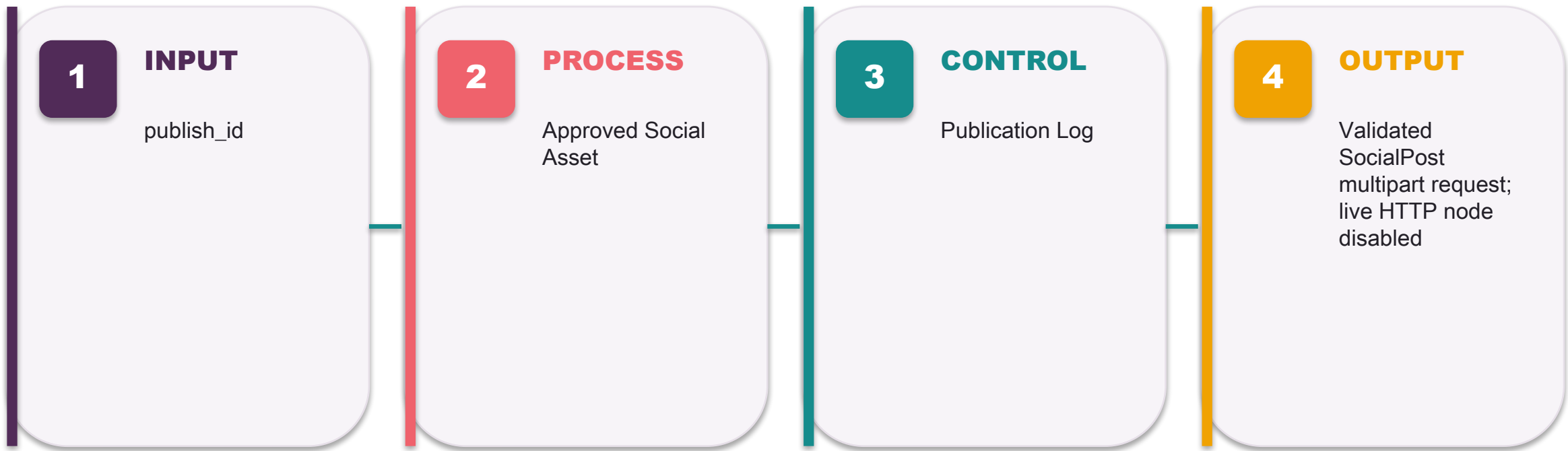


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 11 — Observable Evidence



The inspected request uses the documented SocialPost endpoint and Apikey credential pattern, remains approval-linked and idempotent, and sends nothing unless the trainer explicitly enables the sandbox node.

Lab 11 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Retries can publish the same approved content more than once or to an unapproved SocialPost profile

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: The inspected request uses the documented SocialPost endpoint and Apikey credential pattern, remains approval-linked and idempotent, and sends nothing unless the trainer explicitly enables the sandbox node.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 11 — From Slide Model to Lab Evidence



Lab 12 — End-to-End Campaign Orchestrator

S

Scenario

Build campaign stage 12 of the connected Northstar Launch marketing automation.

A

Artifact

A parent workflow that calls each stage, records state and stops safely on failure

I

Input

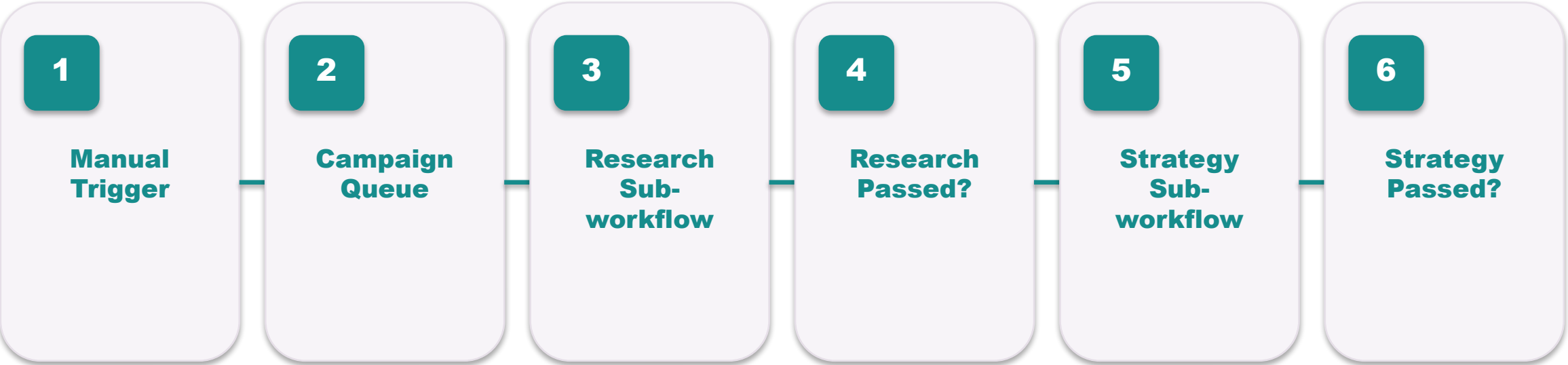
data/orchestrator_runs.xlsx · 8 named fields

✓

Acceptance

A successful run reaches SOCIALPOST_REQUEST_INSPECTED only after research, strategy, content, QA and approval succeed; failures stop downstream execution.

Lab 12 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 12 — input and output contract

FIELD 01 run_id	FIELD 02 campaign_id
FIELD 03 stage	FIELD 04 status
FIELD 05 started_at	FIELD 06 ended_at
FIELD 07 evidence_uri	FIELD 08 error_code

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 12 — state and ownership



Lab 12 — metric and decision rule

$$\text{stage_success_rate} = \text{successful_stages} / \text{attempted_stages}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{stage_success_rate} = \text{successful_stages} / \text{attempted_stages}$

Lab 12 — failure containment



Failure mode

A parent agent may continue after a failed or unapproved stage

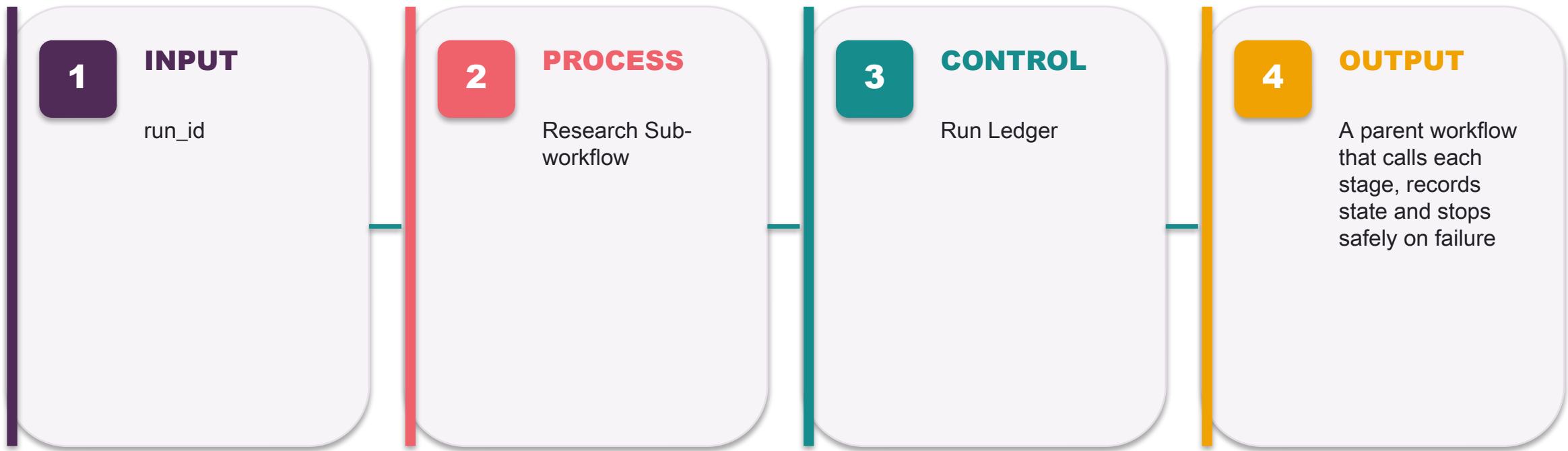


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 12 — Observable Evidence



A successful run reaches **SOCIALPOST_REQUEST_INSPECTED** only after research, strategy, content, QA and approval succeed; failures stop downstream execution.

Lab 12 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. A parent agent may continue after a failed or unapproved stage

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets:
A successful run reaches SOCIALPOST_REQUEST_INSPECTED only after research, strategy, content, QA and approval succeed; failures stop downstream execution.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 12 — From Slide Model to Lab Evidence



TOPIC 05

Performance Analytics & Optimisation

Event ingestion · attribution · ROI · anomaly detection · controlled feedback

LO2 · K2/K4 · A3-A4

Architecture · runtime · contract · controls · evidence



Event contract

M

Mechanism

campaign_id, channel, event, value, cost and timestamp join delivery to outcomes.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Metric grain

M

Mechanism

Calculate at campaign × channel × period before rolling up totals.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Incrementality

M

Mechanism

Optimisation
recommendations distinguish
correlation from controlled lift.

C

Concrete contract

Named fields, owner,
timestamp, version and
acceptance status make this
mechanism testable.

!

Failure control

Reject missing lineage,
ambiguous ownership or
state transitions that exceed
delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Guardrails

M

Mechanism

CPA, frequency, unsubscribe rate and brand-risk thresholds can override ROAS.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Drift detection

M

Mechanism

Schema, volume and conversion-rate changes trigger investigation—not automatic action.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

Learning loop

M

Mechanism

Evidence proposes a change; a human approves; the next experiment measures impact.

C

Concrete contract

Named fields, owner, timestamp, version and acceptance status make this mechanism testable.

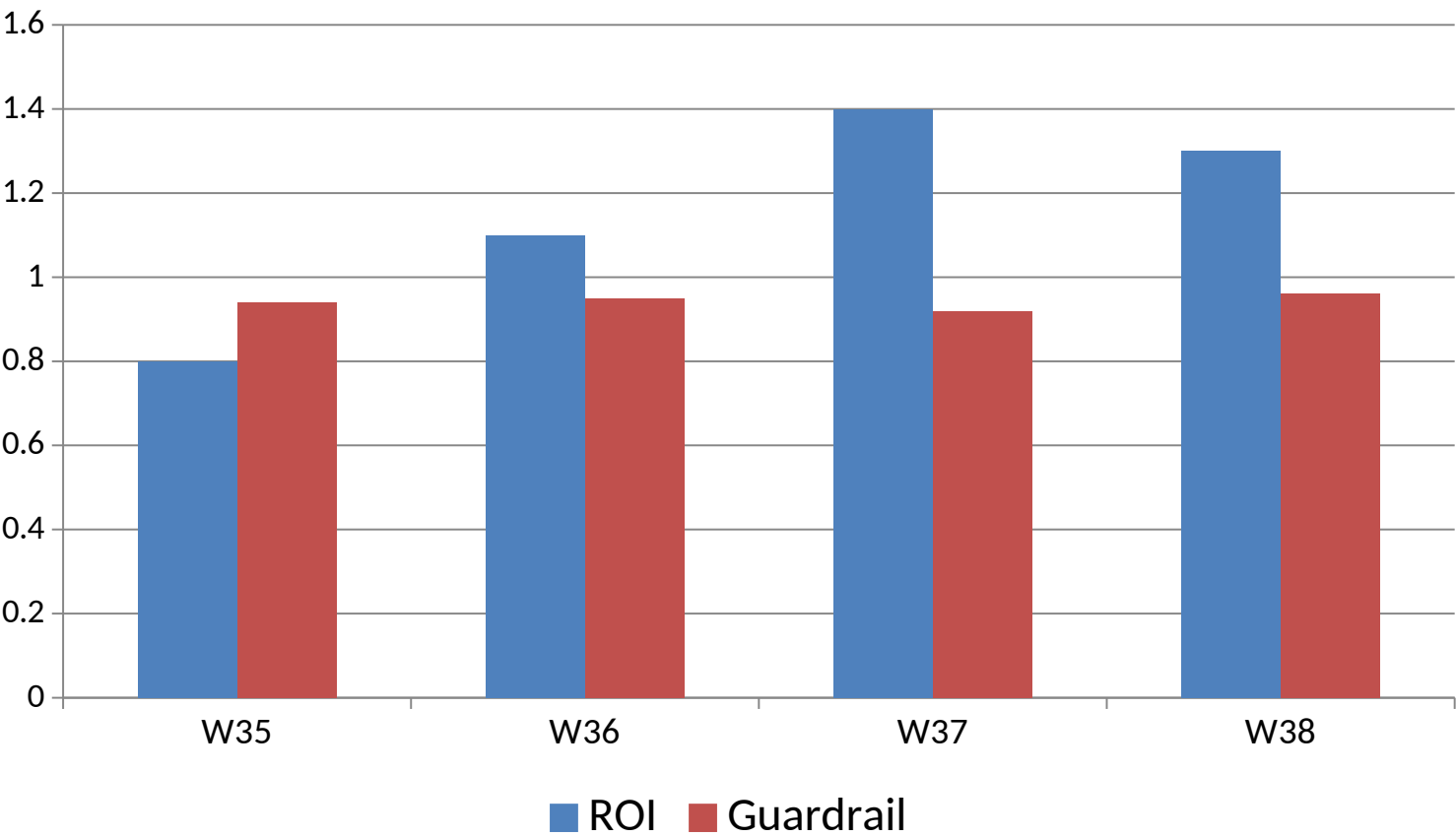
!

Failure control

Reject missing lineage, ambiguous ownership or state transitions that exceed delegated authority.

Practice bridge: inspect the Excel contract and n8n execution evidence.

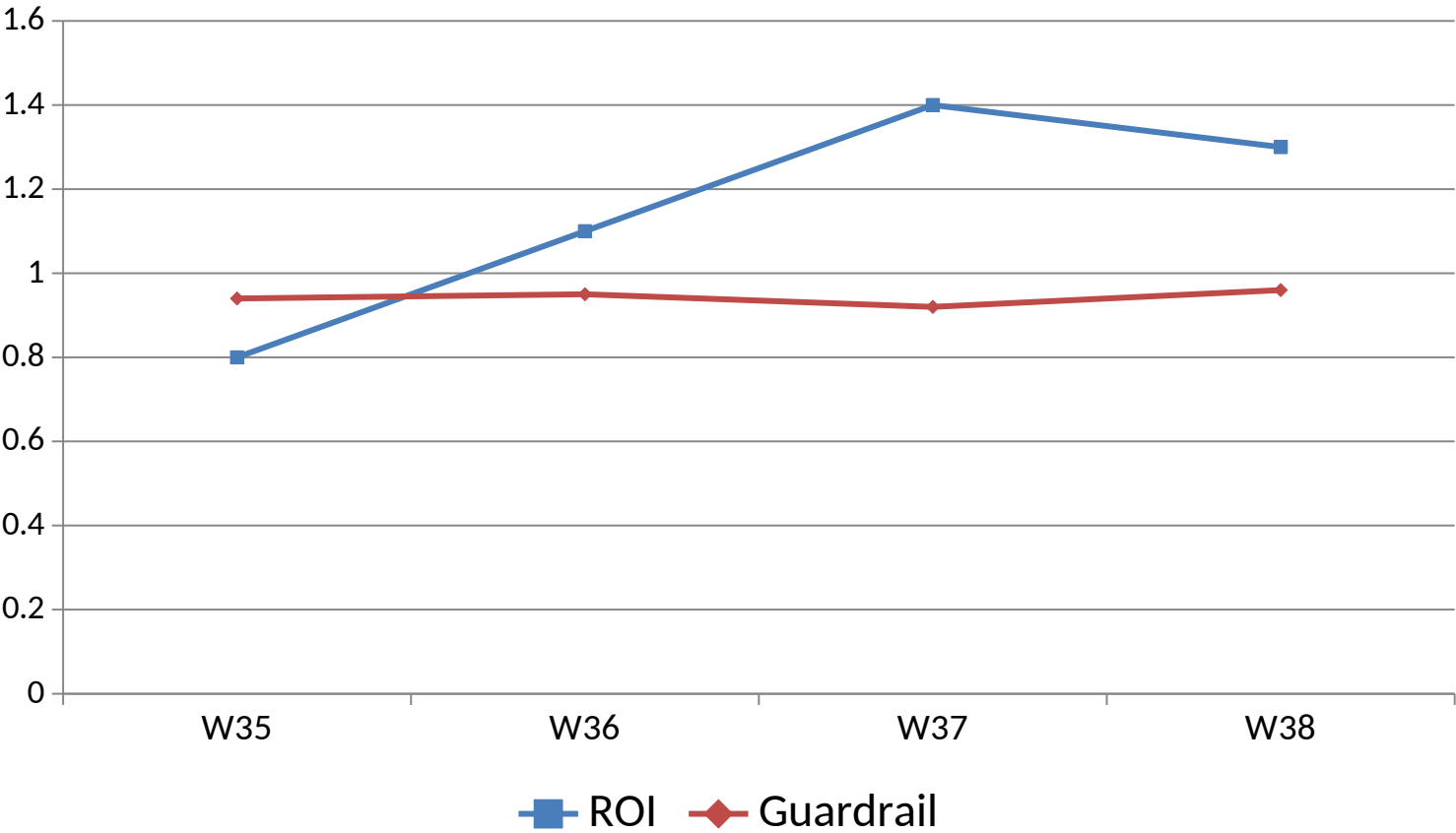
Topic 05 — worked dataset



What the data shows

Optimisation considers return and guardrail health; one spike is not causal lift.

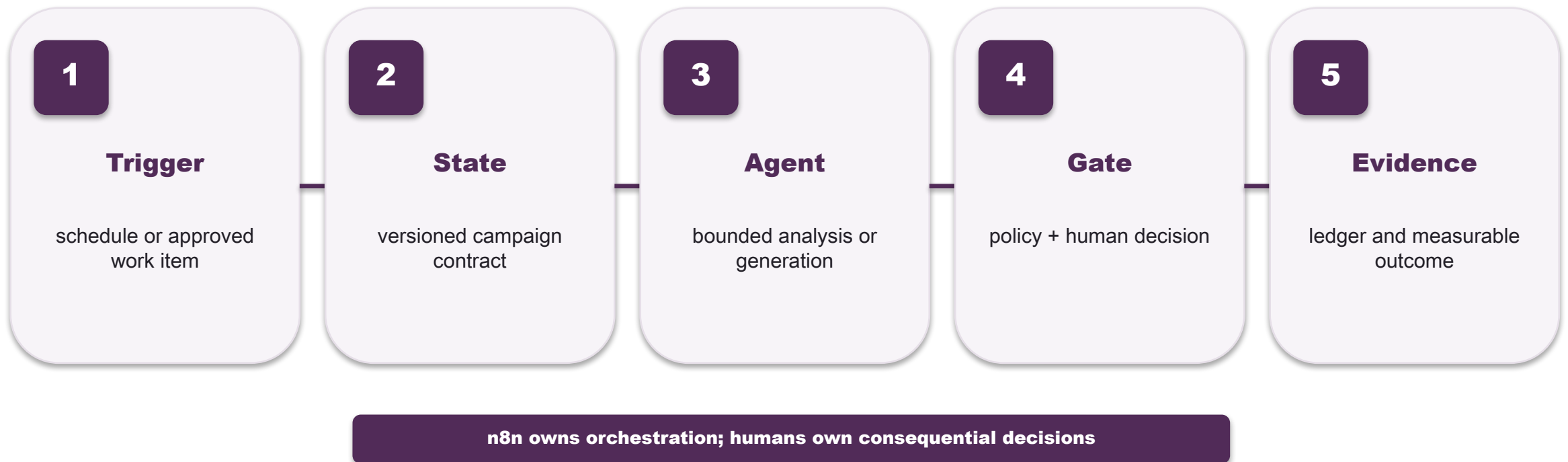
Topic 05 — trend and decision threshold



What the data shows

Read direction, volume and guardrails together; record what would change the decision.

Topic 05 — Control-Plane Architecture



Topic 05 — Control Case

1

Input conflict

Two plausible signals support different actions; one has stronger lineage and lower risk.

2

Decision rule

Prefer reproducible evidence, explicit assumptions and reversibility. Escalate unclear authority or risk.

3

Evidence

Record chosen input, rejected alternative, owner, threshold and next review date.

Lab 13 — Performance Event Ingestion & Attribution

S

Scenario

Build campaign stage 13 of the connected Northstar Launch marketing automation.

A

Artifact

A deduplicated fact table joined by campaign, asset, channel and event time

I

Input

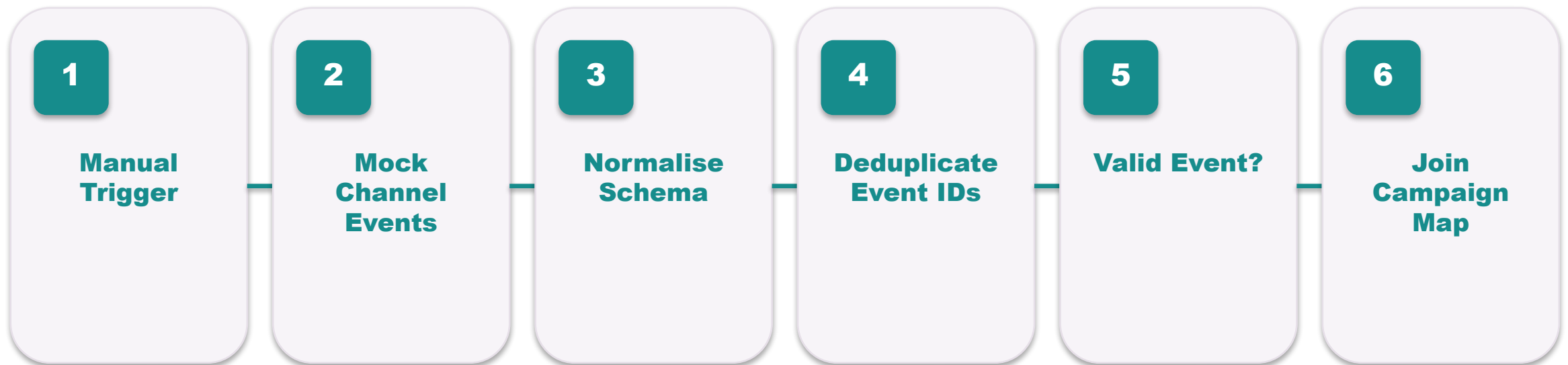
data/performance_events.xlsx · 9 named fields

✓

Acceptance

All facts have unique event_id, valid campaign_id, normalised time and an explicit attribution rule or quarantine reason.

Lab 13 — n8n execution path



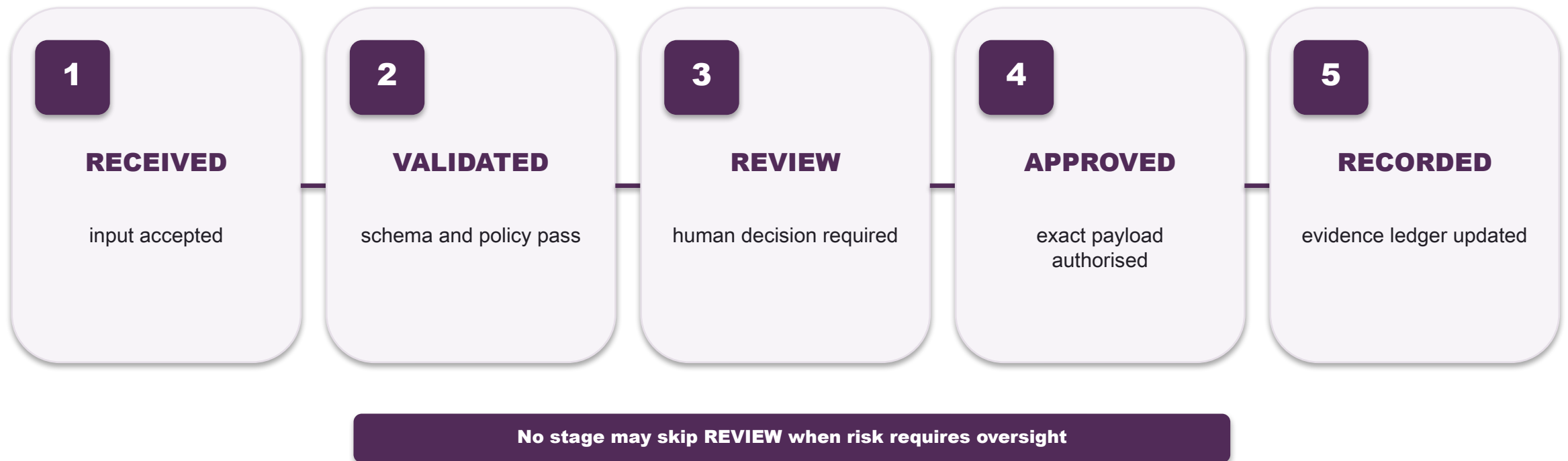
Detailed build procedure is in the Learner Guide

Lab 13 — input and output contract

FIELD 01 event_id	FIELD 02 campaign_id
FIELD 03 asset_id	FIELD 04 channel
FIELD 05 event	FIELD 06 event_time
FIELD 07 value_sgd	FIELD 08 cost_sgd

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 13 — state and ownership



Lab 13 — metric and decision rule

$$\text{data_quality} = \text{valid_unique_events} / \text{received_events}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{data_quality} = \text{valid_unique_events} / \text{received_events}$

Lab 13 — failure containment



Failure mode

Inconsistent IDs and time zones can misattribute outcomes

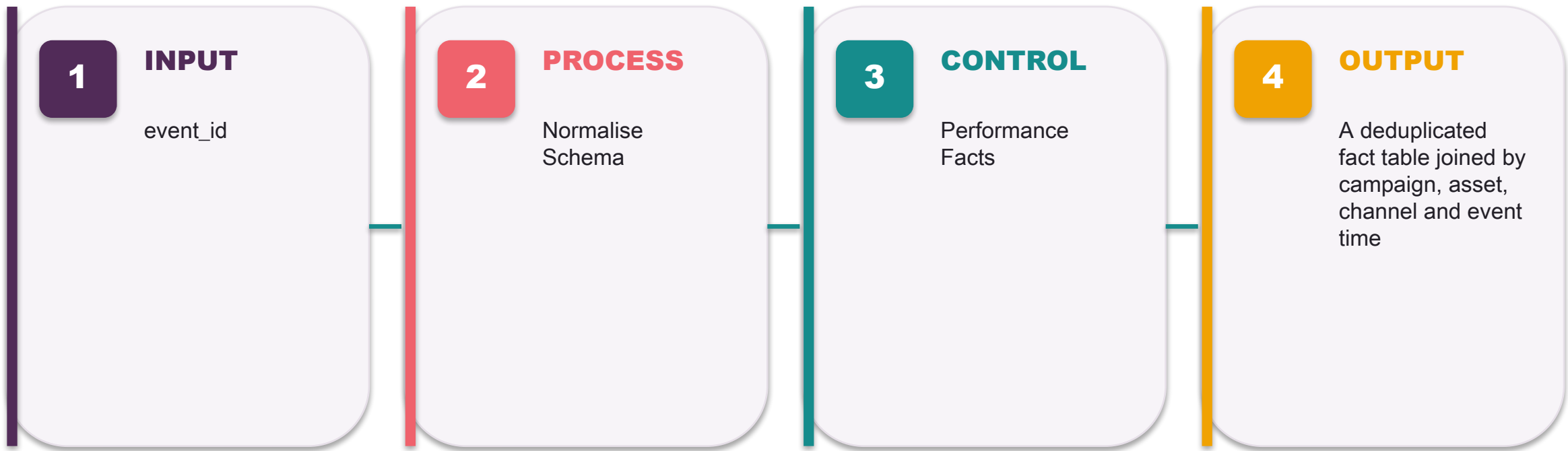


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 13 — Observable Evidence



All facts have unique event_id, valid campaign_id, normalised time and an explicit attribution rule or quarantine reason.

Lab 13 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Inconsistent IDs and time zones can misattribute outcomes

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets:
All facts have unique event_id, valid campaign_id, normalised time and an explicit attribution rule or quarantine reason.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 13 — From Slide Model to Lab Evidence



Lab 14 — ROI Dashboard & Anomaly Detection

S

Scenario

Build campaign stage 14 of the connected Northstar Launch marketing automation.

A

Artifact

A channel-period metric table with CAC, ROAS, ROI, confidence and anomaly flags

I

Input

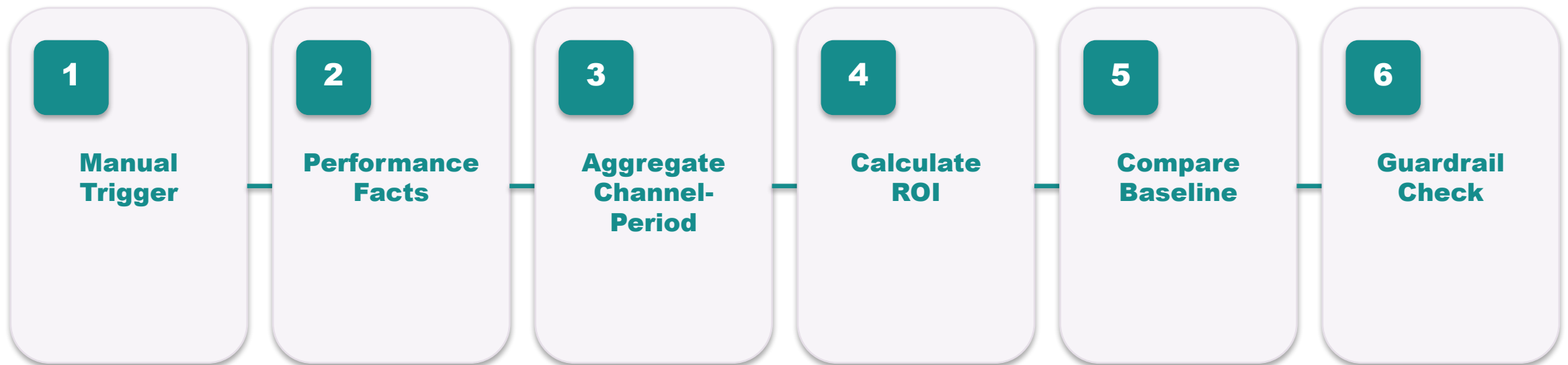
data/roi_dashboard.xlsx · 8 named fields

✓

Acceptance

Dashboard metrics match the formulas; anomalies include volume, baseline, guardrail status and analyst decision.

Lab 14 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 14 — input and output contract

FIELD 01 period	FIELD 02 channel
FIELD 03 spend	FIELD 04 customers
FIELD 05 revenue	FIELD 06 gross_margin
FIELD 07 baseline_cv	FIELD 08 current_cv

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 14 — state and ownership



Lab 14 — metric and decision rule

$$\text{ROI} = (\text{revenue} \times \text{gross_margin} - \text{spend}) / \text{spend}; \text{ lift} = (\text{current} - \text{baseline}) / \text{baseline}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: $\text{ROI} = (\text{revenue} \times \text{gross_margin} - \text{spend}) / \text{spend}$; lift = $(\text{current} - \text{baseline}) / \text{baseline}$

Lab 14 — failure containment



Failure mode

Small samples can produce dramatic but unreliable lift

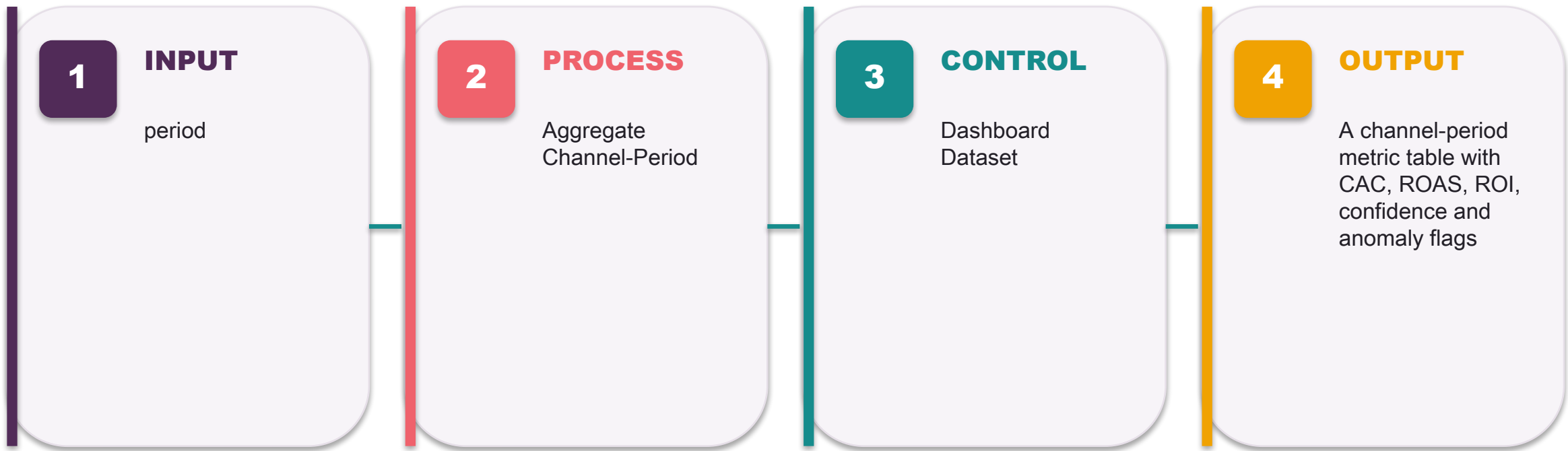


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 14 — Observable Evidence



Dashboard metrics match the formulas; anomalies include volume, baseline, guardrail status and analyst decision.

Lab 14 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. Small samples can produce dramatic but unreliable lift

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets: Dashboard metrics match the formulas; anomalies include volume, baseline, guardrail status and analyst decision.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

Lab 14 — From Slide Model to Lab Evidence



Lab 15 — Optimisation Feedback Loop & Capstone

S

Scenario

Build campaign stage 15 of the connected Northstar Launch marketing automation.

A

Artifact

A controlled recommendation, experiment update and approved next-cycle campaign plan

I

Input

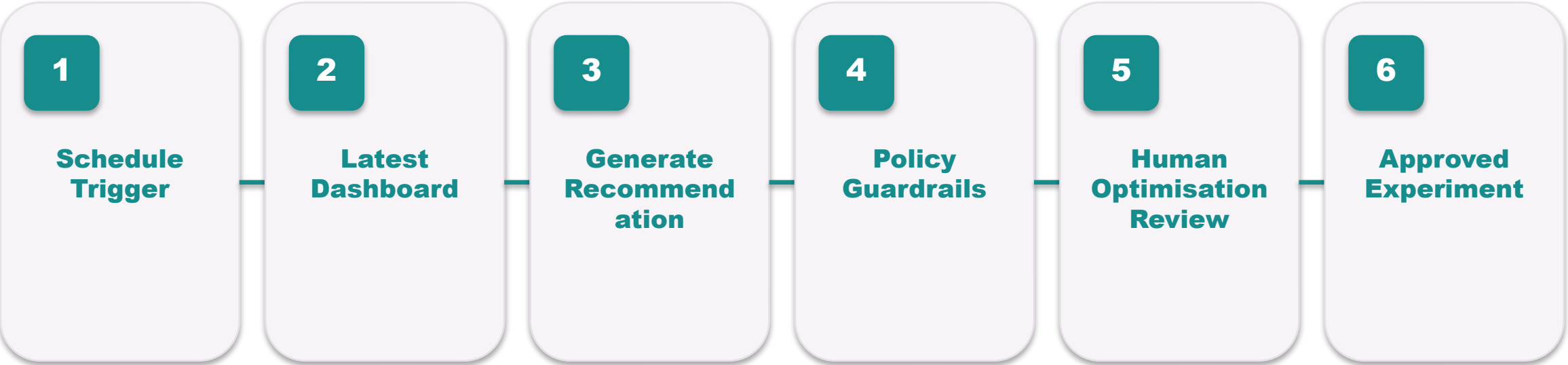
data/optimisation_actions.xlsx · 9 named fields

✓

Acceptance

No optimisation changes a live campaign automatically; approved recommendations create a measured next-cycle experiment with full lineage.

Lab 15 — n8n execution path



Detailed build procedure is in the Learner Guide

Lab 15 — input and output contract

FIELD 01 action_id	FIELD 02 finding
FIELD 03 evidence	FIELD 04 proposed_change
FIELD 05 expected_impact	FIELD 06 risk
FIELD 07 owner	FIELD 08 decision

Every downstream node reads this contract; schema drift routes to quarantine.

Lab 15 — state and ownership



Lab 15 — metric and decision rule

$$\text{realised_lift} = (\text{new_result} - \text{baseline}) / \text{baseline}$$



Worked example

Use the supplied mock workbook; reproduce the metric and compare with n8n output. Preserve units, grain and zero-division handling.



Decision rule

Accept only when reproducible and evidence meets the threshold. Metric: realised_lift = (new_result - baseline) / baseline

Lab 15 — failure containment



Failure mode

A self-optimising agent can chase short-term metrics and exceed delegated authority

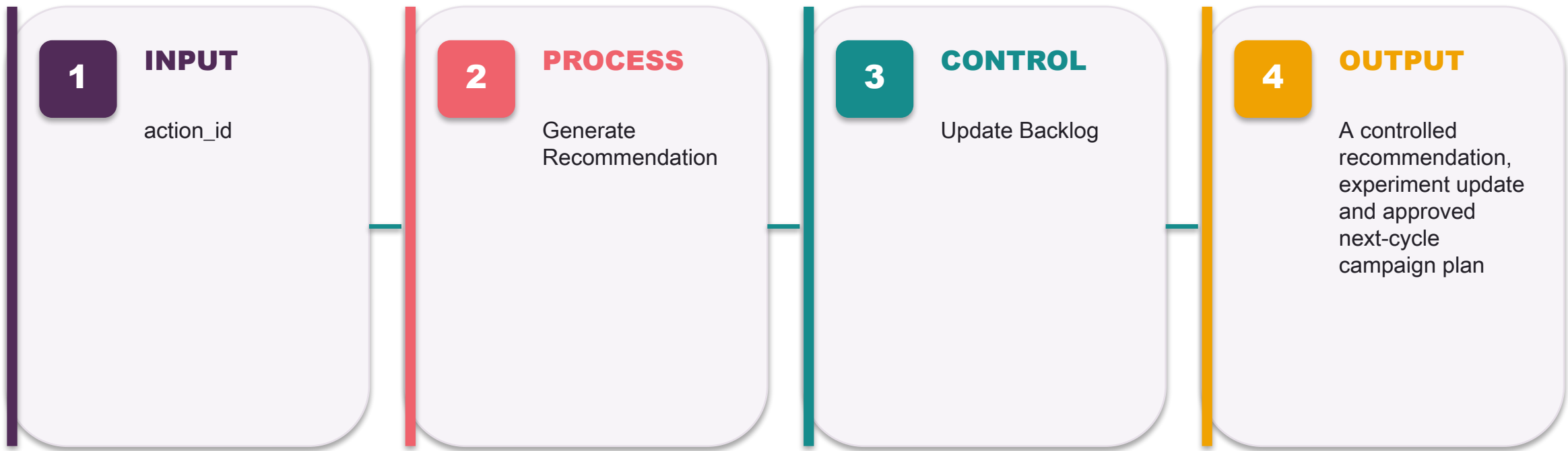


Control and recovery

Validate the contract, stop downstream execution, preserve the failed payload, assign an owner and require evidence before retry.

Automation stops safely; a named owner decides the next action.

Lab 15 — Observable Evidence



No optimisation changes a live campaign automatically; approved recommendations create a measured next-cycle experiment with full lineage.

Lab 15 — Decision Under Uncertainty

1

Observed signal

The output is plausible, but one assumption has weak evidence. A self-optimising agent can chase short-term metrics and exceed delegated authority

2

Human decision

Inspect lineage, compare the workbook, and approve only the exact version that meets:
No optimisation changes a live campaign automatically;
approved recommendations create a measured next-cycle experiment with full lineage.

3

Recorded outcome

Store reviewer, decision, timestamp, payload hash, evidence URI and next review date.

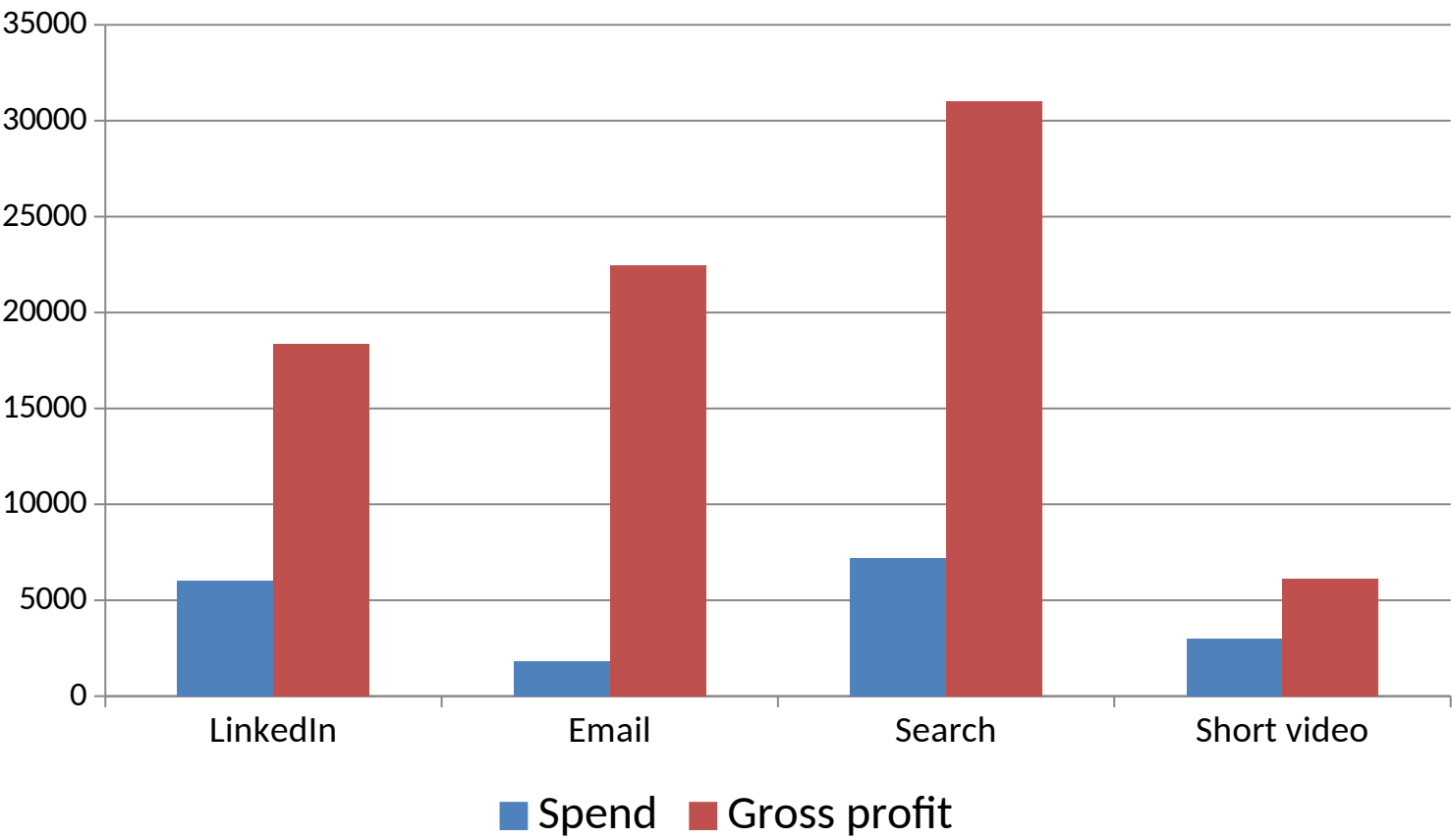
Lab 15 — From Slide Model to Lab Evidence



Capstone — Research to Social Publishing



Capstone campaign economics



What the data shows

Email and Search lead on contribution, but incrementality and volume still govern the next decision.

Capstone stop conditions



Failure mode

Missing evidence, schema drift, rejected QA, payload hash mismatch, unavailable reviewer or channel error.



Control and recovery

Stop downstream execution, write the failed state, preserve payload evidence and assign a human owner.

Automation stops safely; a named owner decides the next action.

What You Achieved

1

Research

Evidence-led research with citation lineage.

2

Strategy

RACE/SOSTAC, KPI contracts and ROI decisions.

3

Operations

Content, QA, approval and dry-run publishing.

4

Optimisation

Measured feedback without autonomous live mutation.

Assessment Reminder



Explain

Use mechanisms, fields, formulas and controls.



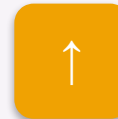
Build

Use the supplied workflow and mock Excel data.



Verify

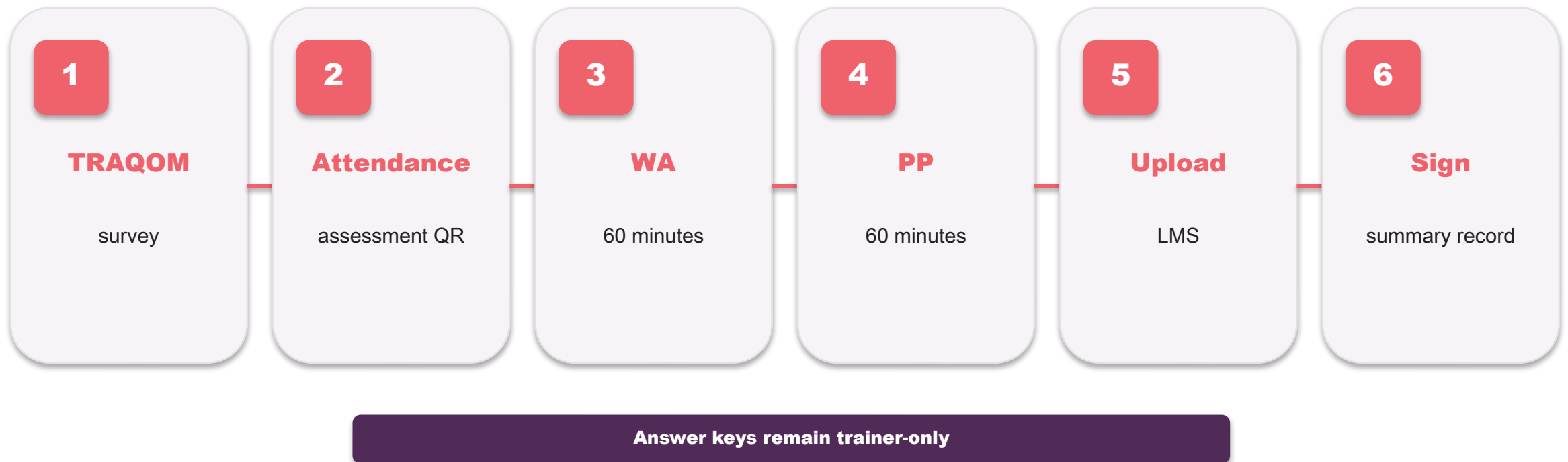
Show execution evidence and checks.



Submit

Upload question papers through the LMS.

Assessment Flow



Digital Attendance (Mandatory)

PM

PM attendance

Confirm final training attendance.

A

Assessment attendance

Scan the dedicated assessment QR.

T

TRAQOM

Complete course feedback.

✓

Check

Show confirmation if requested.

Thank You

Automate the workflow. Preserve human judgement.

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